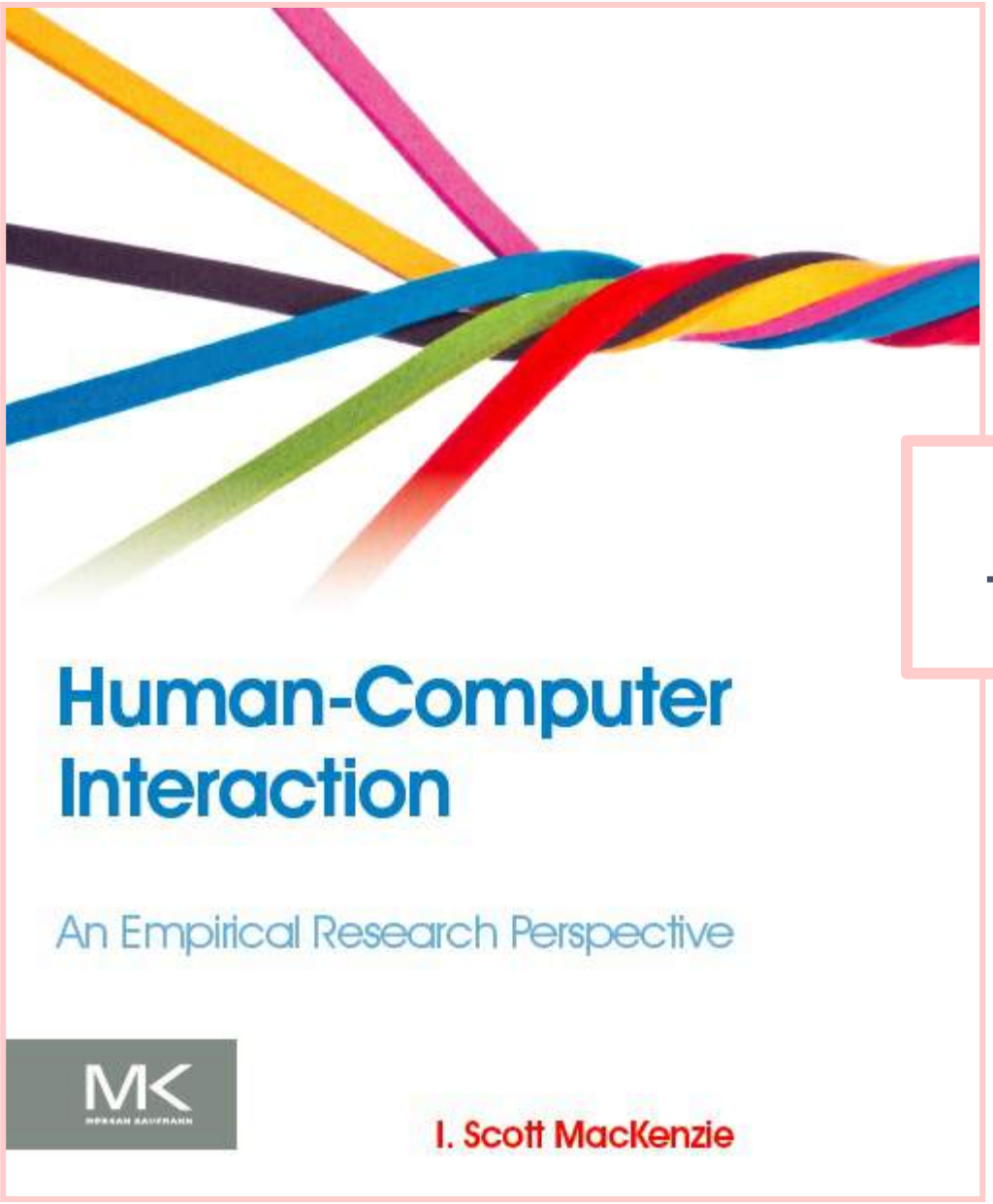


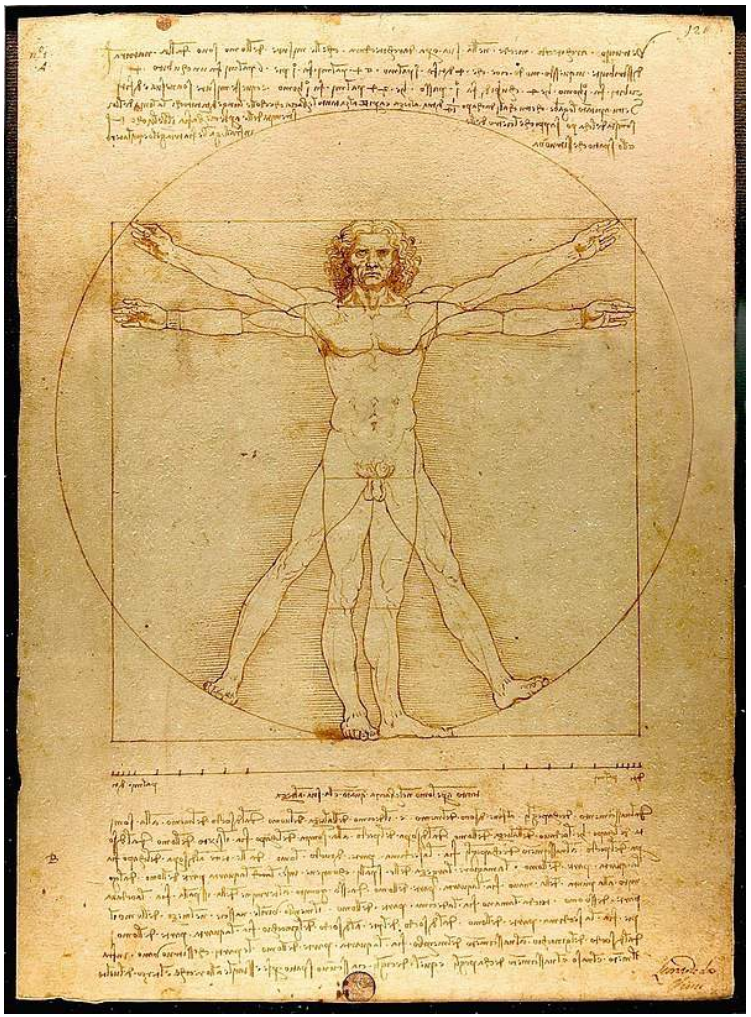
Chapter 2
The Human Factor

Longbiao CHEN
2018/04 Xiamen University

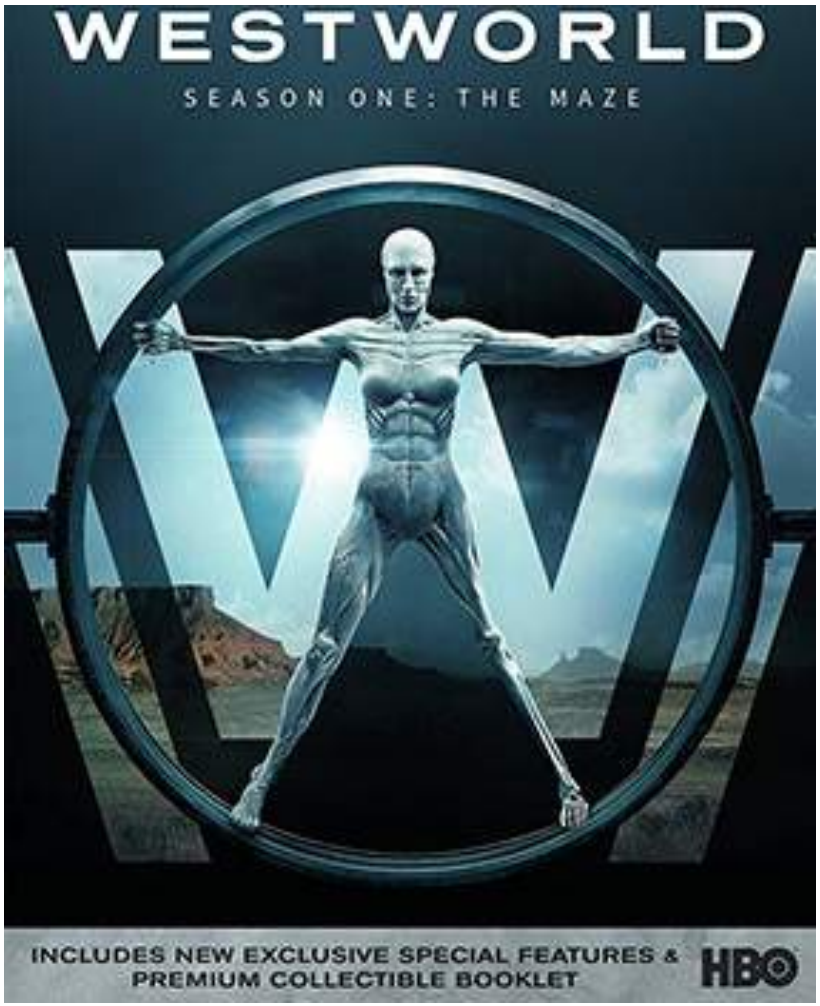
Models of the Human

- **Descriptive** models are tools for thinking
- Useful to have a descriptive model for the human
- There are many models of human

Models of the Human

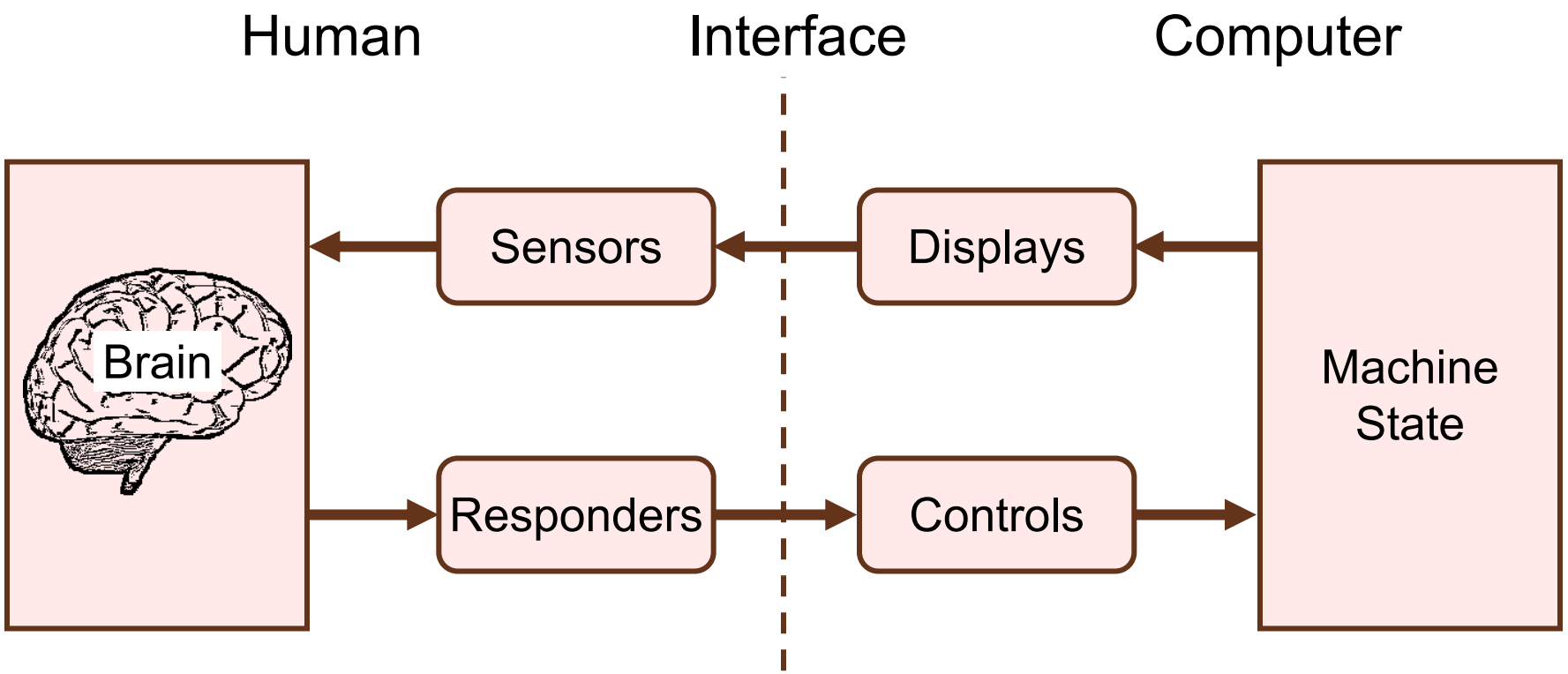


Leonardo da Vinci, *Vitruvian Man*, 1490



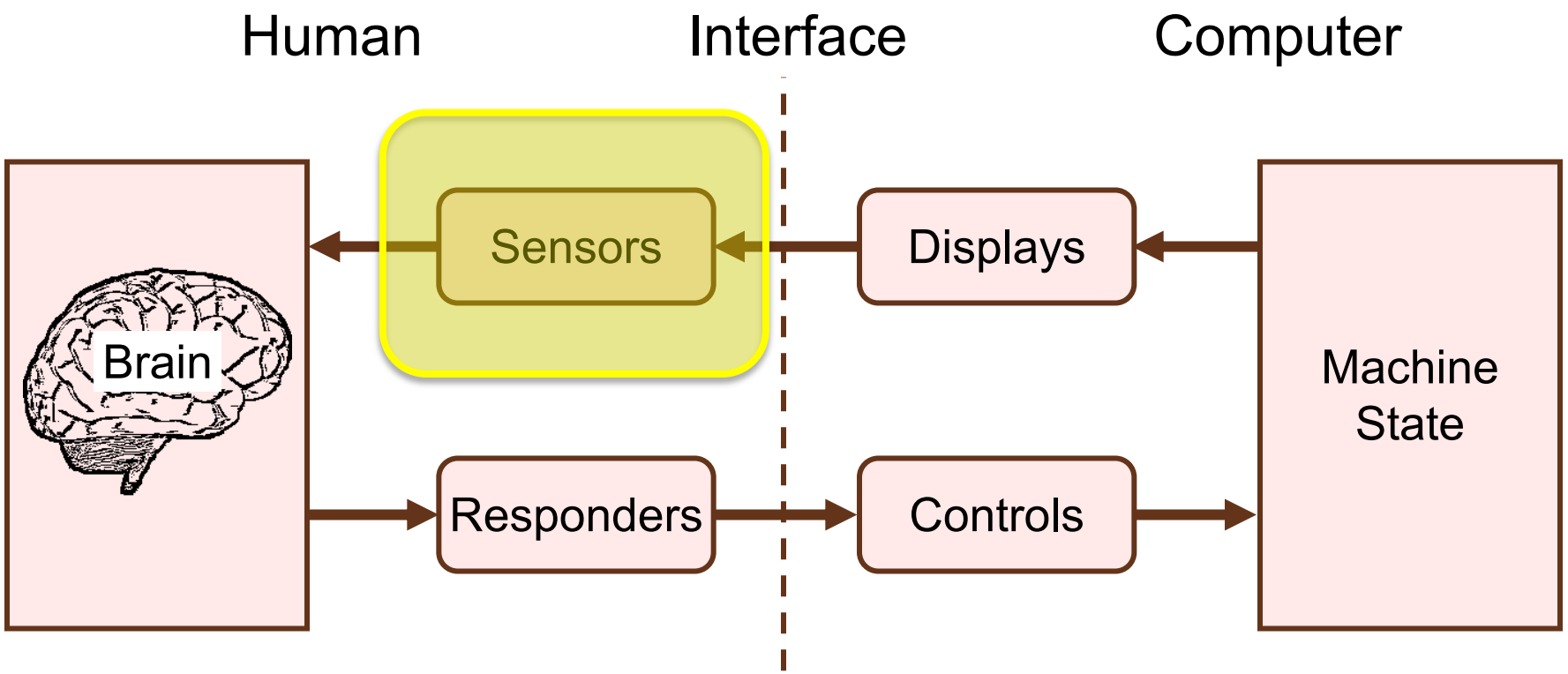
Westworld, *Season 1*, 2017

Human Factors Model[1]



[1] Kantowitz, B. H., & Sorkin, R. D. (1983). *Human factors: Understanding people-system relationships*. New York. New York: Wiley.

Human Factors Model[1]



[1] Kantowitz, B. H., & Sorkin, R. D. (1983). *Human factors: Understanding people-system relationships*. New York. New York: Wiley.

Human Senses

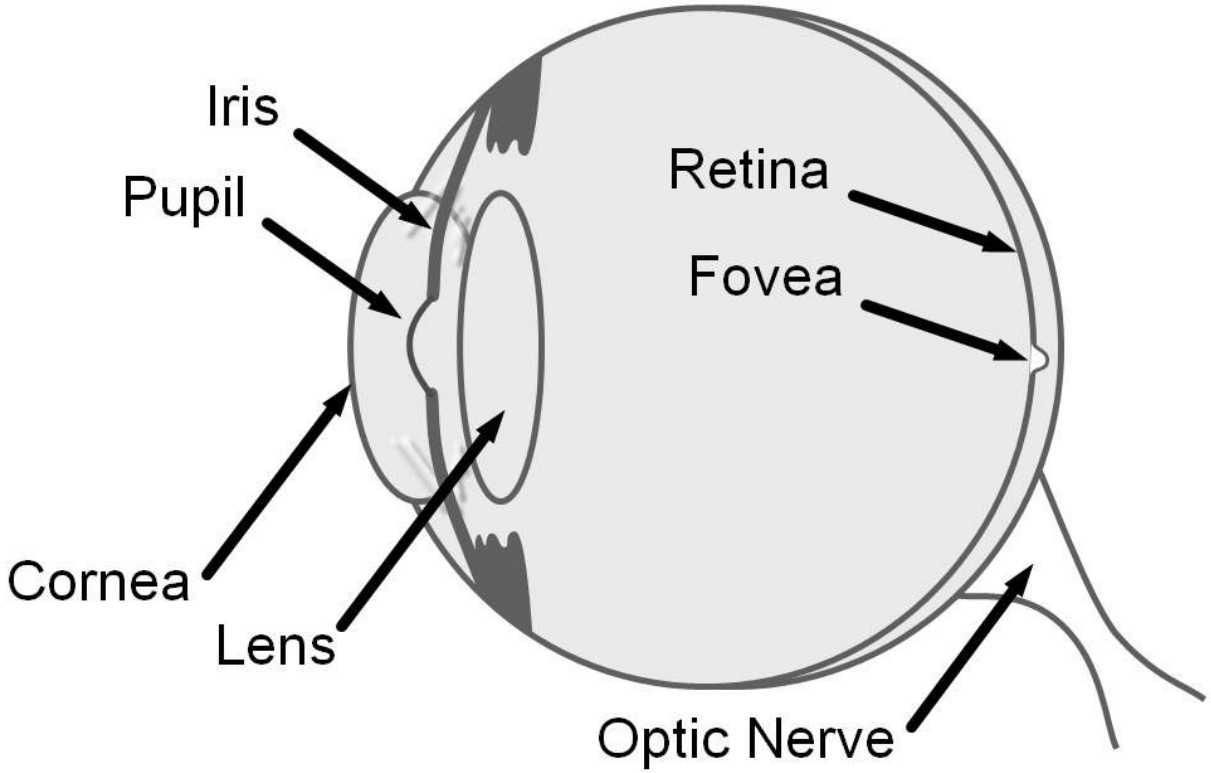
Rosa: You deny everything except what you want to believe. That's the sort of man you are.
Bjartur: I have my five senses, and don't see what need there is for more.
(Halldór Laxness, *Independent People*)

- The five senses: [五识]
  - Vision (sight) [眼，色]
  - Hearing (audition) [耳，声]
  - Touch (tactition) [身，触]
  - Smell (olfaction) [鼻，香]
  - Taste (gustation) [舌，味]



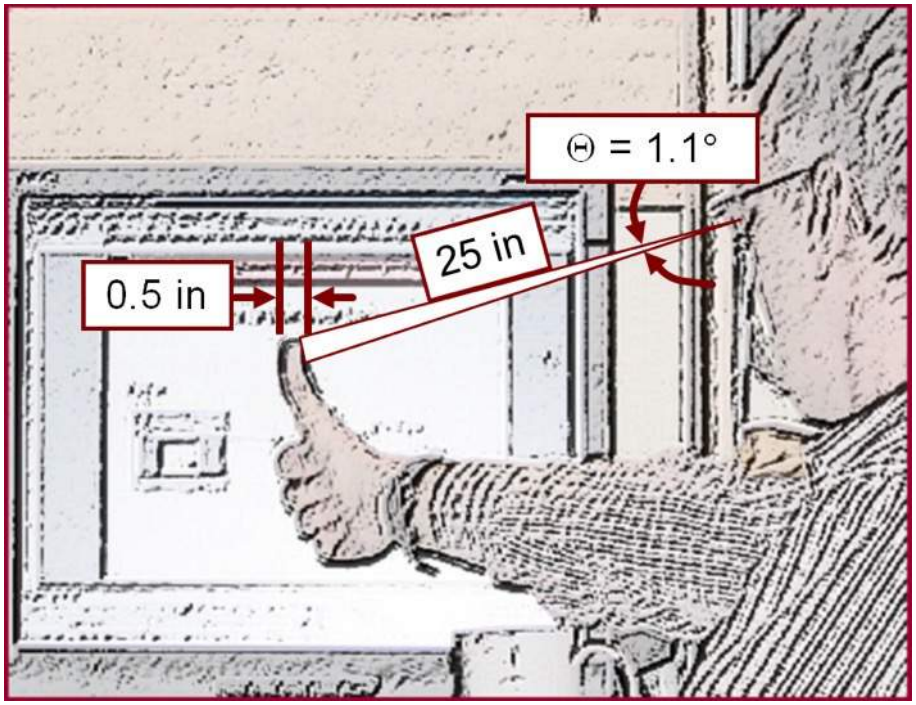
Vision (The Eye)

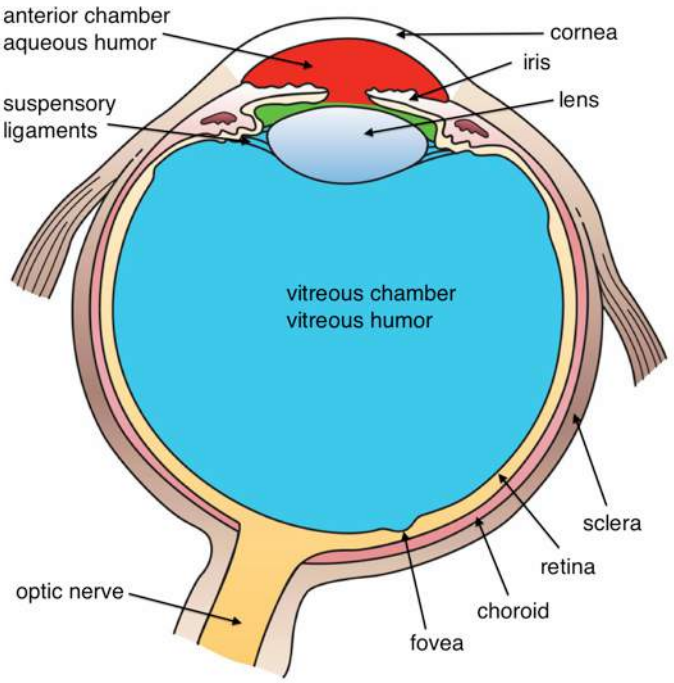
- People obtain about 80% of their information through vision (the eye)



Fovea Image

- Sharp central vision [The Fovea of the Retina]
- 1% of retina, 50% of visual cortex
- Fovea image is ≈1° of visual angle:

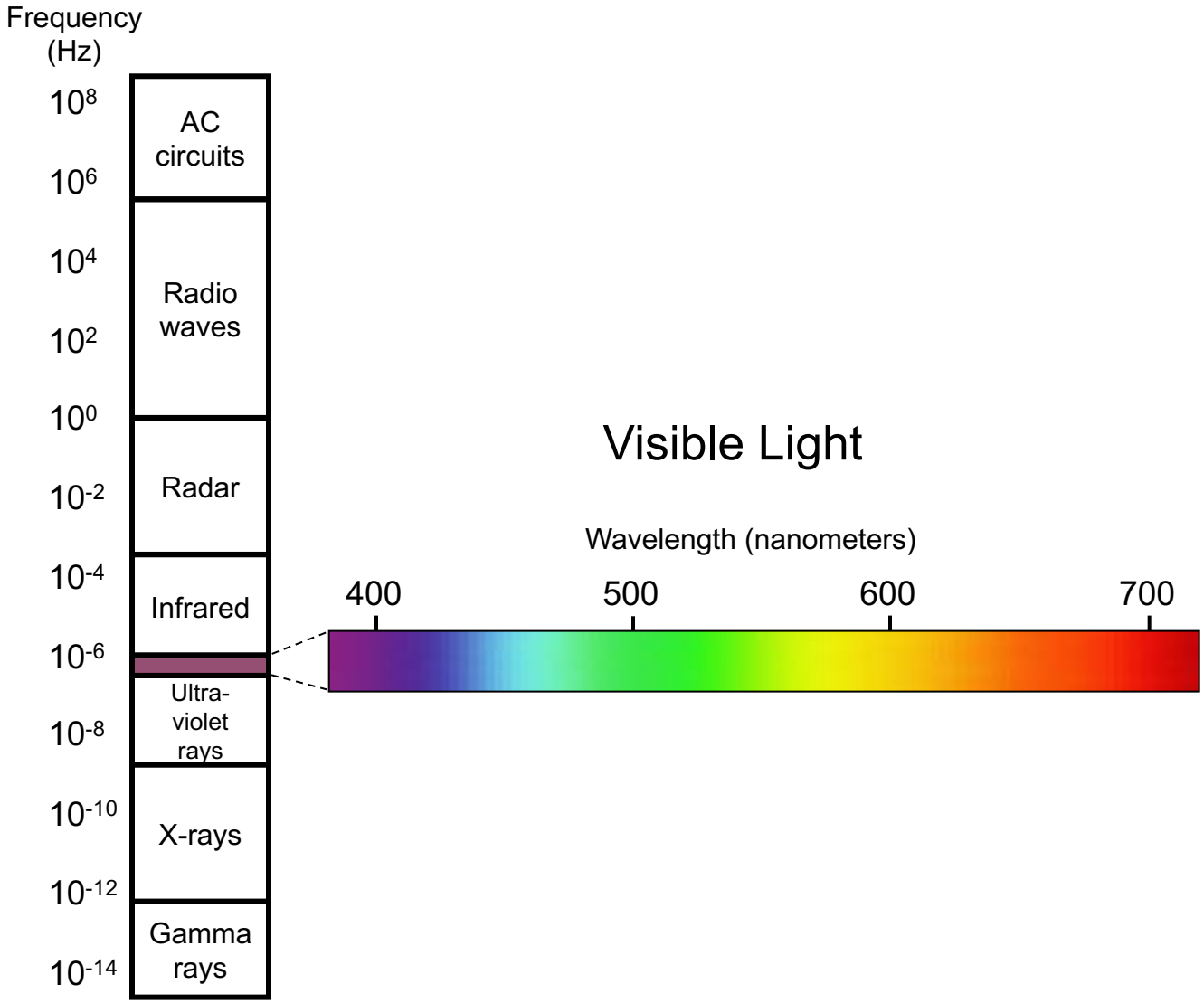




Visual Stimulus

- Physical properties of light…
  - Frequency
  - Intensity (luminance)
- Create subjective properties of vision…
  - Color (next slide)
  - Brightness

Colour Spectrum



Fixations and Saccades

- **Fixation** [注视]
  - Eyes are stationary (dwell) [停留]
  - Take in visual detail from the environment
  - Long or short, but typically at least **200 ms**
- **Saccade** [扫视]
  - Rapid repositioning of the eye to fixate on a new location
  - Quick: ≈120 ms



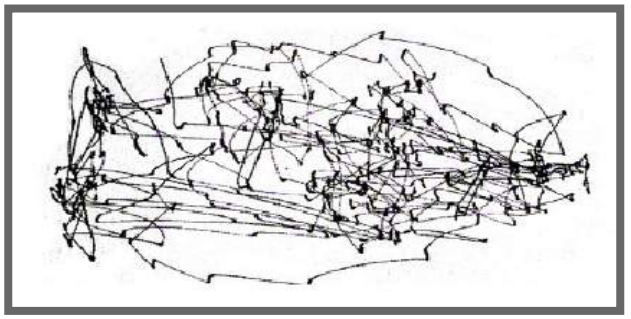
[Tobii eye tracker]

[Eye tracker fixation saccades in flight simulator]

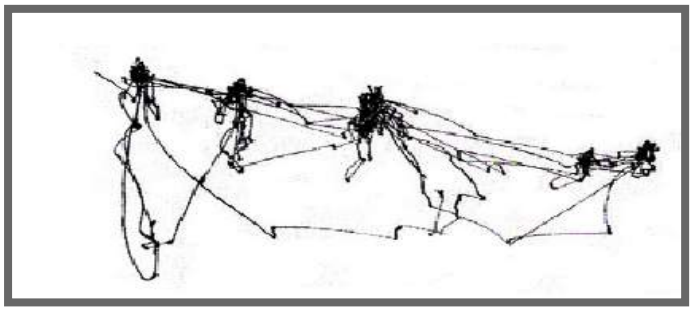
Yarbus' Eye Tracking Research (1965)[1]



The Unwanted Visitor
by Ilya Repin (1844-1930)



"Remember the position of people and objects in the room"



"Estimate the ages of the people"

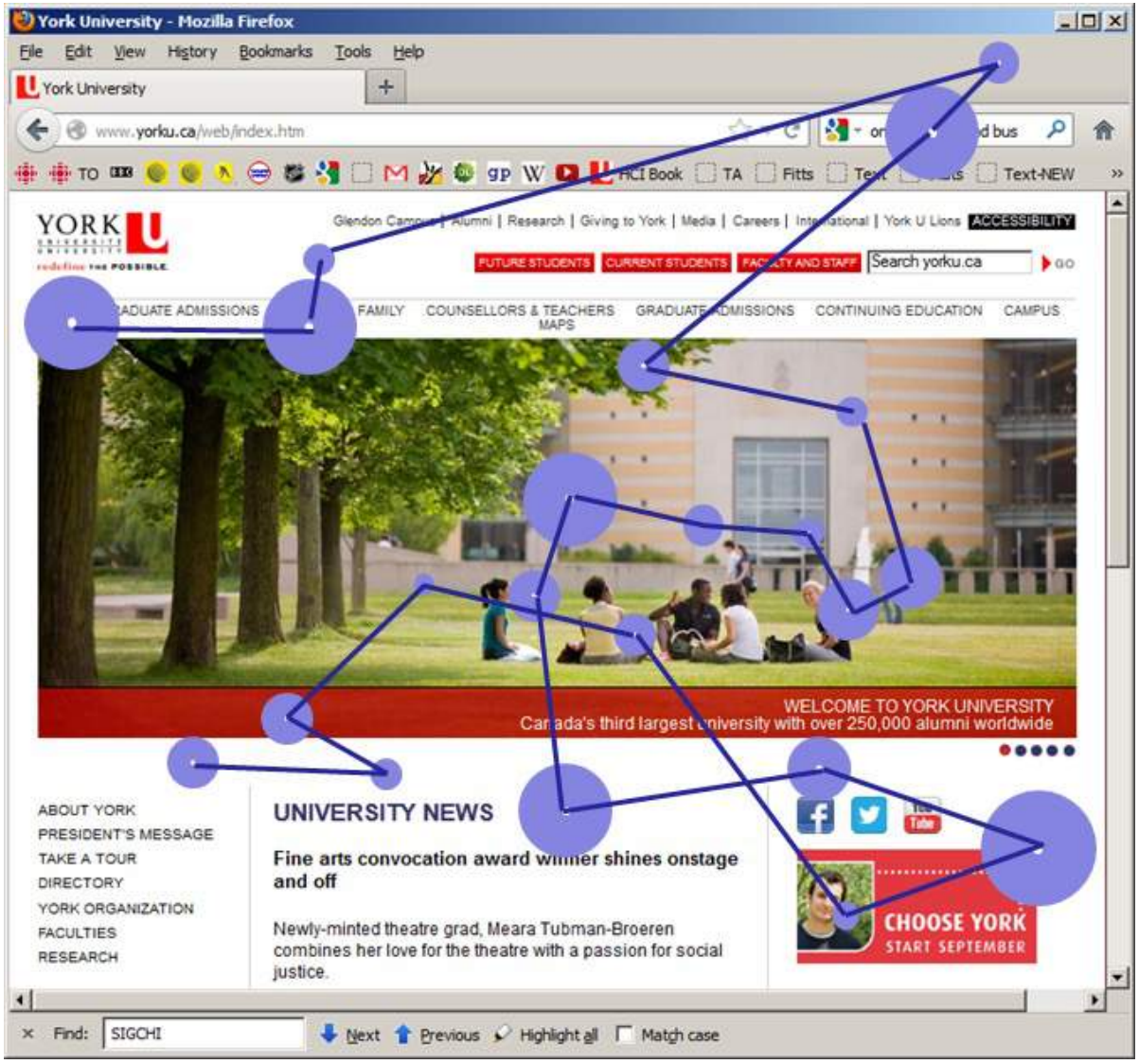
[1] Tatler, B. W., Wade, N. J., Kwan, H., Findlay, J. M., & Velichkovsky, B. M. (2010). Yarbus, eye movements, and vision. *i-Perception, 1*, 7-27..

Scan Paths

- Visual depiction of saccades and fixations
  - Saccades → straight lines
  - Fixations → circles
  - Diameter of circle ∝ duration of fixation
- Applications
  - User behaviour research (e.g., reading patterns)
  - Marketing research (e.g., ad placement)

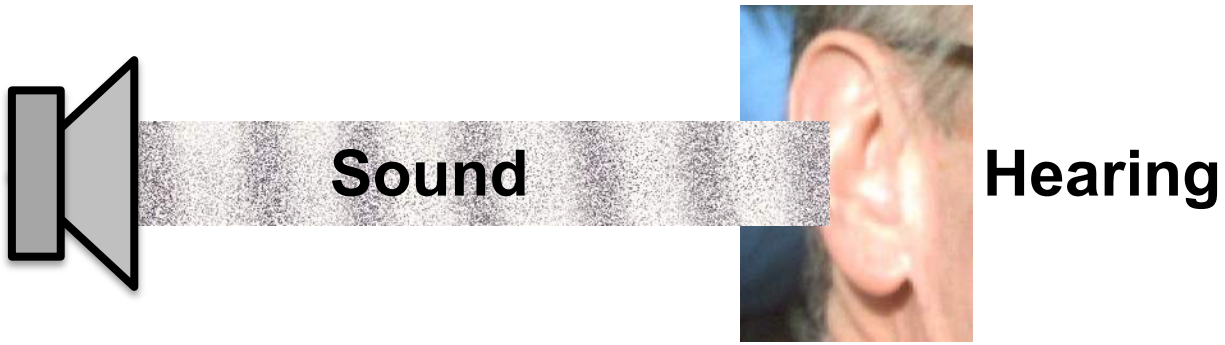
[The next generation PC Gaming Eye Tracking Device]

Scan Path Example



Hearing (Audition)

- Sound: cyclic fluctuations of pressure in a medium
  - e.g., air, water
- Created when physical objects are moved or vibrated
- Examples
  - Slamming a door, plucking a guitar string, shuffling cards, speaking (via larynx)



Auditory Stimulus

- Physical properties of sound…
  - Frequency [频率]
  - Intensity [强度]
- Create subjective properties of hearing…
  - Pitch [声调]
  - Loudness [响度]
  - Timbre (next slide) [音色]
  - Attack (after next slide) [起音]

[Pitch and Loudness of Sound Waves]

Timbre



- Aka *richness*, *brightness, tone color*
- Results from harmonic structure of sound
- E.g., a musical note of 200 Hz, has harmonics at 400 Hz, 600 Hz, 800 Hz, etc.
- Notes of the same frequency from different instruments are distinguished, in part, due to timbre

[Tone Colors]



“trumpet” [小号]

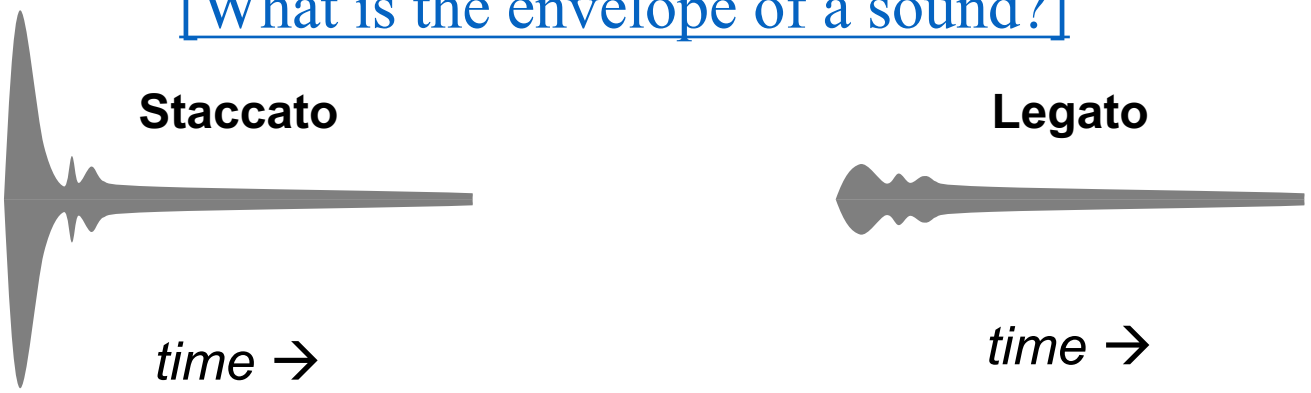


“oboe” [双簧管]

Attack

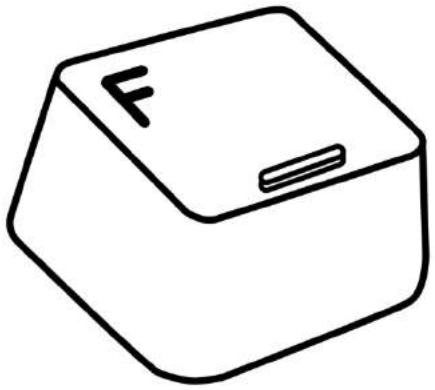
- Aka *envelope: attack, decay, sustain, release*
- Results from the way a note and its harmonics build up and transition in time – from silent, to audible, to silent
- Considerable information in the onset envelop
- Assists in distinguishing notes of the same pitch coming from different instruments
- Onset envelop created through articulation (e.g., legato, staccato)

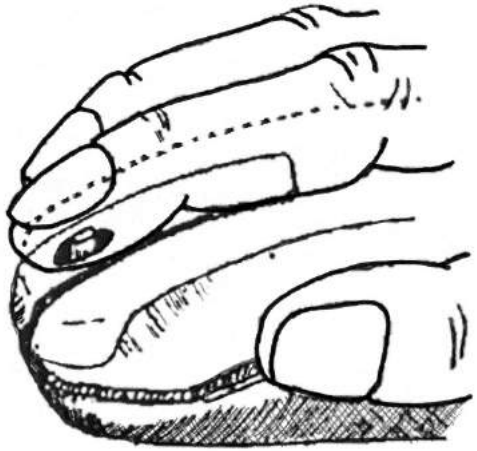
[What is the envelope of a sound?]



Touch (Tactition)

- Receptors in skin, muscles, joints, bones
  - Sense of touch, pain, temperature, position, shape, texture, resistance, etc.
- Tactile feedback examples:







Taptic Feedback

- Tactus keyboard

Buttons when you need them.
Gone when you don't.

Tactus provides a new dimension of user interface with a fundamentally unique solution: application-controlled, completely transparent physical buttons that rise up from the touch-screen surface on demand. With the buttons enabled, users can push and type or rest their fingers as they would with any physical button or keyboard. When the buttons are disabled, they recede into the screen, becoming invisible and leaving a smooth, seamless flat touch-screen with maximum viewing area.



http://tactustechnology.com/technology/

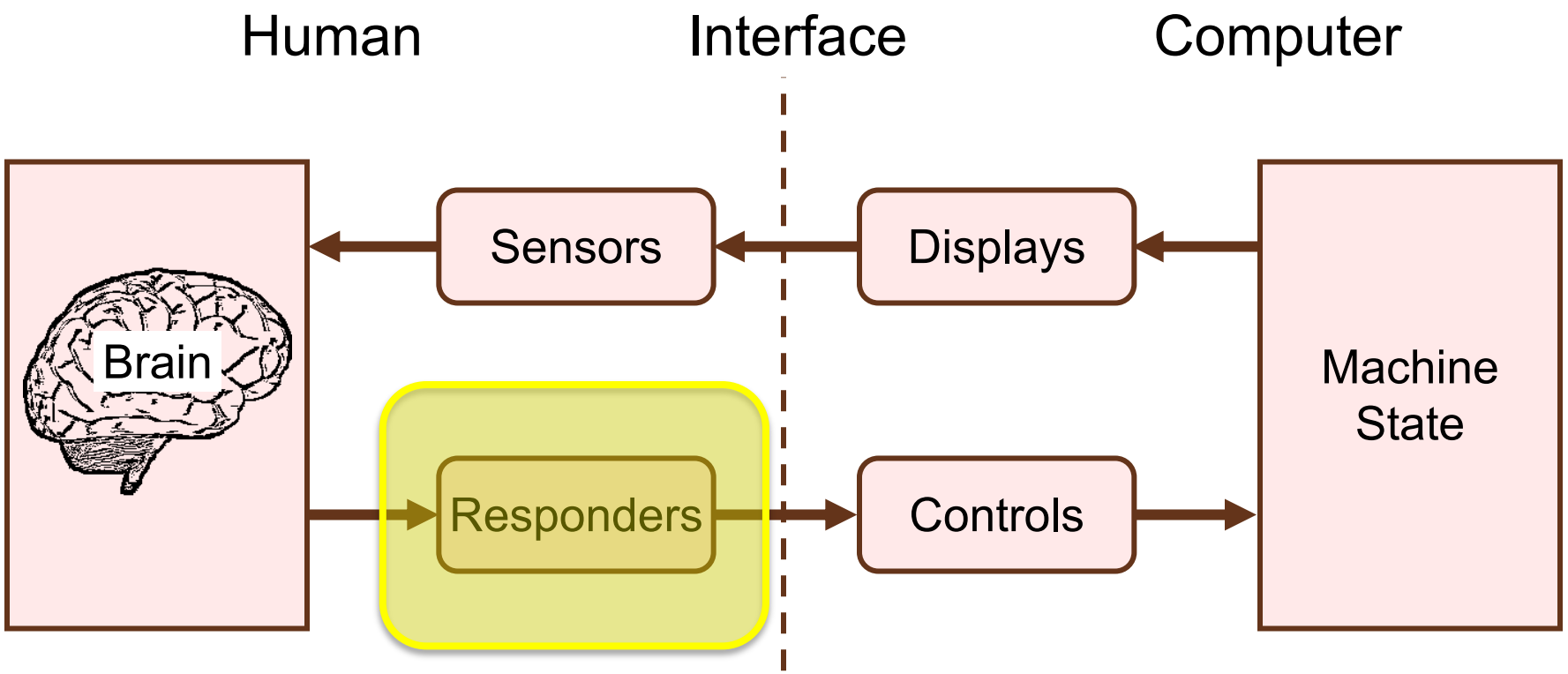
[Apple Taptic Engine]

Smell and Taste

- Smell (olfaction)
  - Ability to perceive odours [气味]
  - Occurs through sensory cells in nasal cavity [鼻孔]
- Taste (gustation)
  - Chemical reception of sweet, salty, bitter, and sour sensations
- Flavour [味道]
  - A perceptual process that **combines** smell and taste
- Only a few examples in HCI (e.g., Brewster et al., 2006; Bodnar et al., 2004)

[Augmented Reality Flavors]

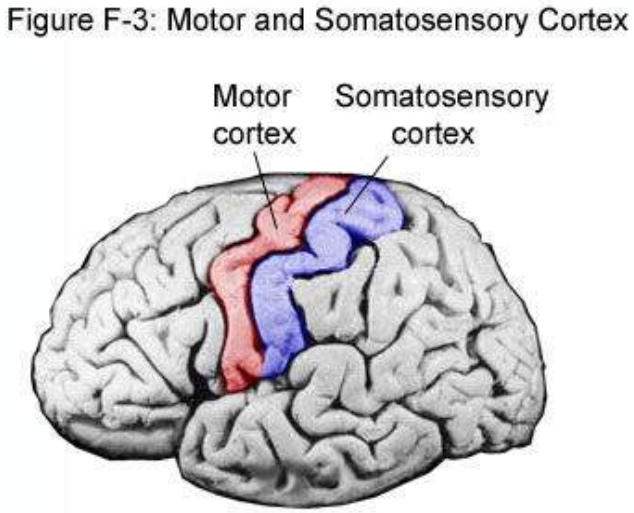
Human Factors Model



Responders

- Humans control their environment through responders, for example…
  - Fingers [手指] to point
  - Feet [脚] to walk or run
  - Eyebrow [眉毛] to frown
  - Vocal chords [声音] to speak
  - Torso [身躯] to lean
- Penfield's (1990) motor homunculus
  - Shows human responders and the relative area of motor cortex dedicated to each (next slide)

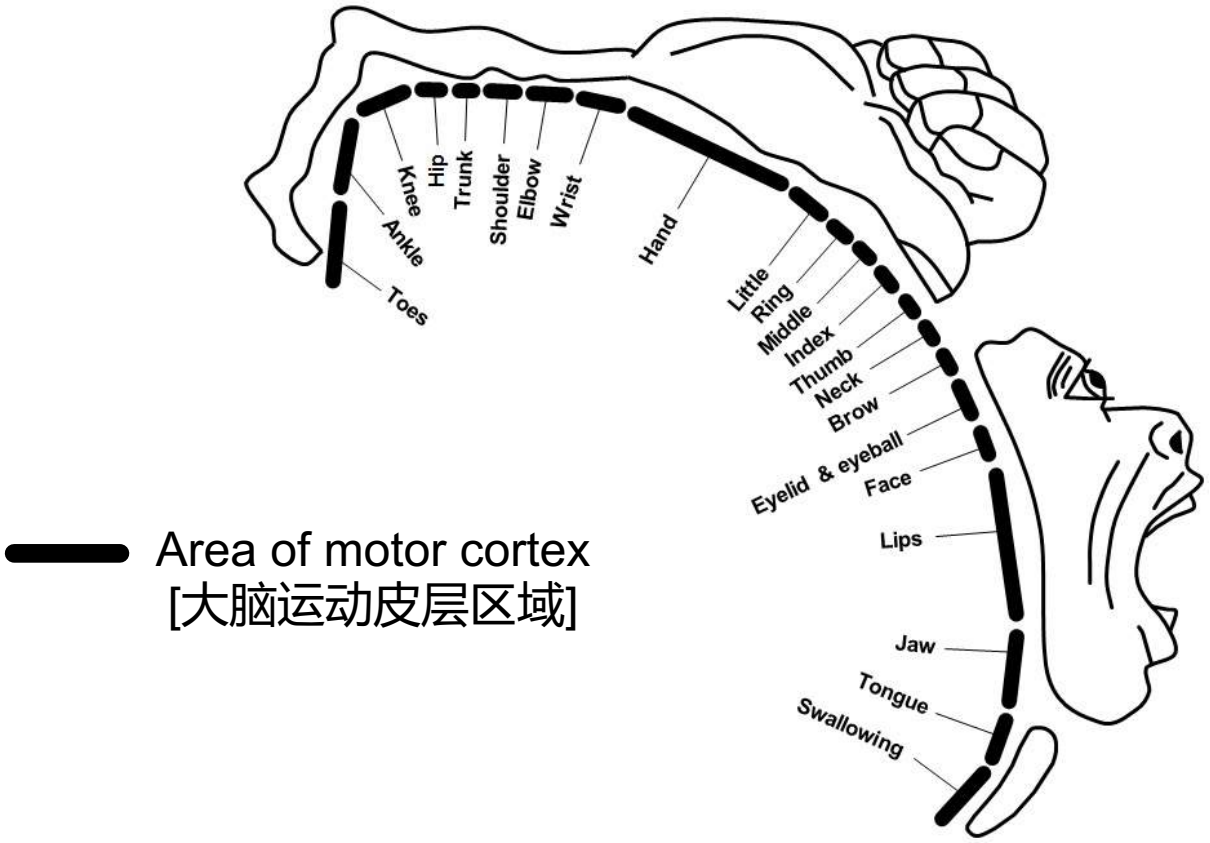
Cortical Homunculus[1][皮质小人]





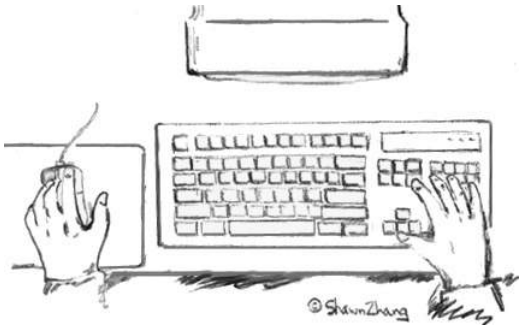
[1] Penfield, W., & Rasmussen, T. (1990). *The cerebral cortex of man: A clinical study of localization of function*. New York: Macmillan.

Motor Homunculus[1][运动小人]



[1] Penfield, W., & Rasmussen, T. (1990). *The cerebral cortex of man: A clinical study of localization of function*. New York: Macmillan.

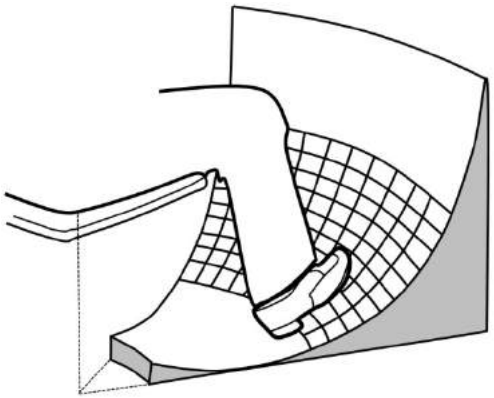
Responder Examples







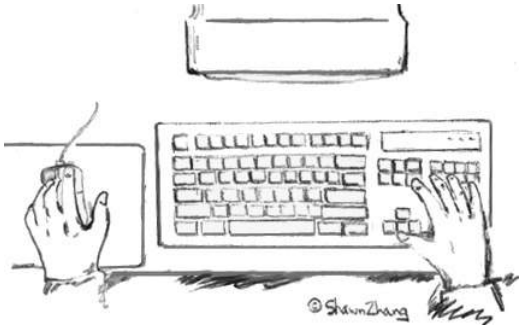






Handedness

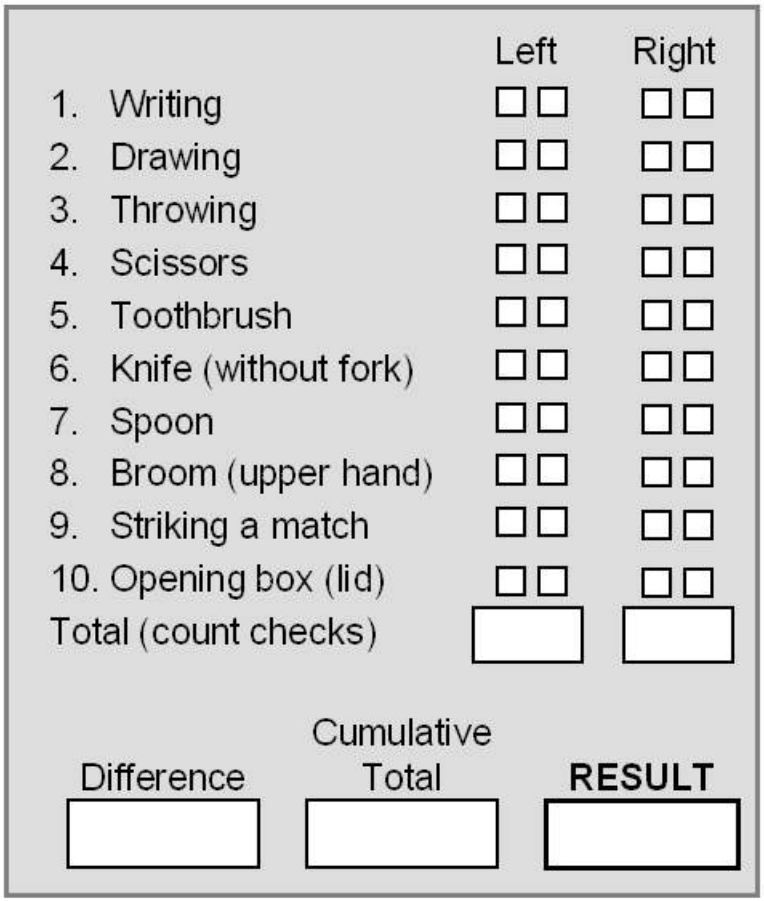
- Some users are left-handed, others right-handed





- Handedness exists by degree
- Edinburgh Handedness Inventory used to measure handedness (next slide)

Edinburgh Inventory for Handedness[1]



	Left	Right
1. Writing	☐☐	☐☐
2. Drawing	☐☐	☐☐
3. Throwing	☐☐	☐☐
4. Scissors	☐☐	☐☐
5. Toothbrush	☐☐	☐☐
6. Knife (without fork)	☐☐	☐☐
7. Spoon	☐☐	☐☐
8. Broom (upper hand)	☐☐	☐☐
9. Striking a match	☐☐	☐☐
10. Opening box (lid)	☐☐	☐☐
Total (count checks)	☐	☐

Difference	Cumulative Total	**RESULT**
☐	☐	☐

Instructions

Mark boxes as follows:

x	preference
xx	strong preference
blank	no preference

Scoring

Add up the number of checks in the "Left" and "Right" columns and enter in the "Total" row for each column. Add the left total and the right total and enter in the "Cumulative Total" cell. Subtract the left total from the right total and enter in the "Difference" cell. Divide the "Difference" cell by the "Cumulative Total" cell (round to 2 digits if necessary) and multiply by 100. Enter the result in the "RESULT" cell.

Interpretation of RESULT

-100 to -40	left-handed
-40 to +40	ambidextrous
+40 to 100	right-handed

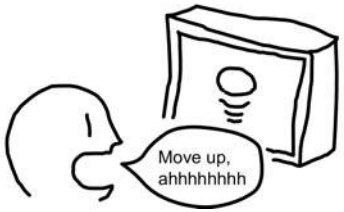
[1] Oldfield, R. C. (1971). The assessment and analysis of handedness: The Edinburgh inventory. *Neuropsychololgia, 9*, 97-113.

Human Voice

- Human vocal chords are responders
- Sounds created through combination of…
  - Movement in the larynx [喉]
  - Pulmonary pressure in the lungs
- Two kinds of vocalized sounds:
  1. Speech
  2. Non-speech
- Both with potential for computer control
  - [Speech + speech recognition] [Apple Siri]
  - Non-speech + signal detection (e.g., frequency, loudness, duration, change direction, etc.)

Non-speech Example[1]

- NVVI = non-verbal voice interaction



	Key 1	Key 2	Key 3	Key 4	BACK
SET 1					
SET 2					
SET 3					
SET 4					

[Yasuhati]

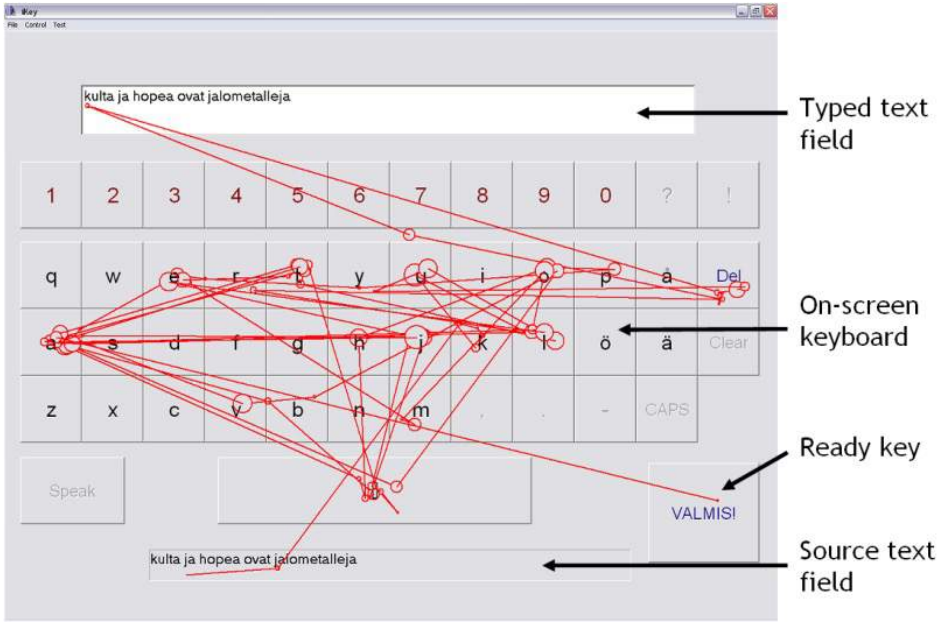
[1] Sporka, A., Felzer, T., Kruniawan, S., Polacek, O., Haiduk, P., & MacKenzie, I. S. (2011). CHANTI: Predictive text entry using non-verbal vocal input. *Proceedings of the ACM Conference on Human Factors in Computing Systems – CHI 2011*, 2463-2472.New York: ACM. (click here)

The Eye as a Responder

- As a responder, the eye is called upon to do "double duty"
  1. Sense and perceive the environment/computer
  2. Act as a controller via saccades and fixations

Example - Eye Typing[1]





[Eye Typing]

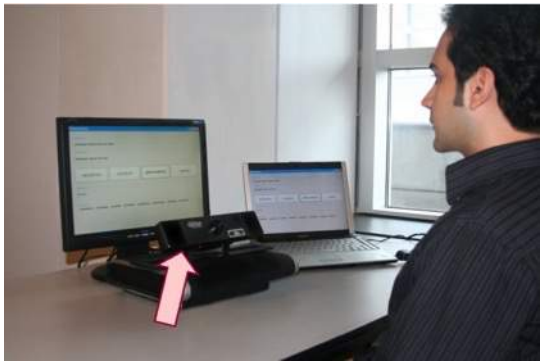
[1] Majaranta, P., MacKenzie, I. S., Aula, A., & Räihä, K.-J. (2006). Effects of feedback and dwell time on eye typing speed and accuracy. *Universal Access in the Information Society (UAIS), 5*, 199-208.

Blinking

- Blinking is an eye response that can be used for computer control:



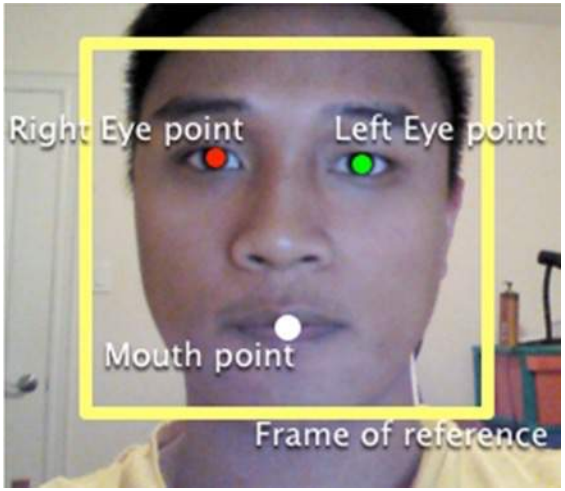
- E.g., BlinkWrite:



[BlinkWrite]

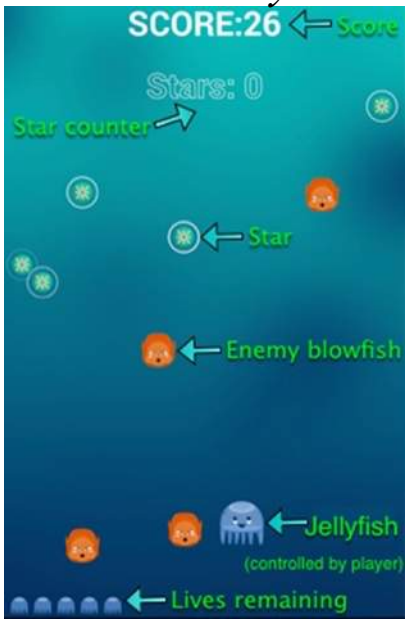
[1] MacKenzie, I. S., & Ashtiani, B. (2011). BlinkWrite: Efficient text entry using eye blinks. *Universal Access in the Information Society*, *10*, 69-80. (click here)

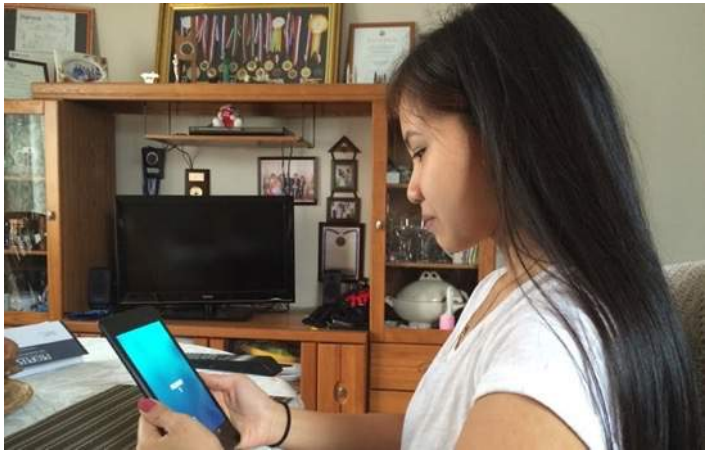
Face Tracking



[Kinect – Facial Tracking]

StarJelly





[iPhone X - Face ID]

Note: The Jellyfish is moved horizontally either by device tilt or by face tracking

[1] Cuaresma, J., & MacKenzie, I. S. (2014). A comparison between tilt-input and facial tracking as input methods for mobile games. *Proc IEEE-GEM 2014*, pp. 70-76, New York: IEEE. (click here)

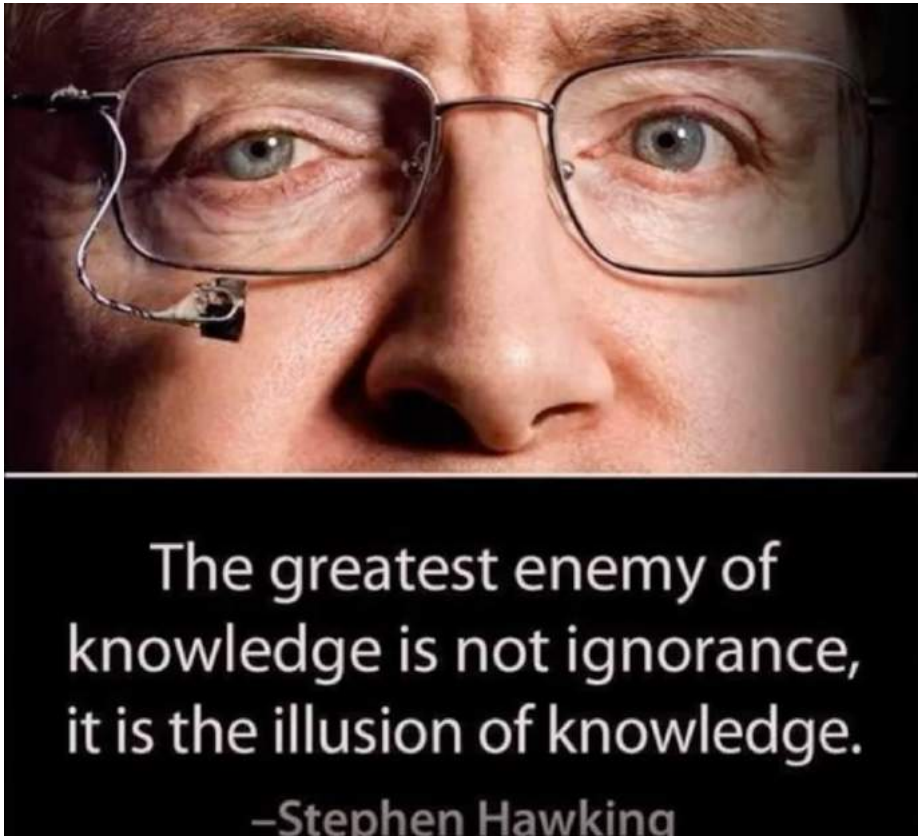
The Sixth Sense Demo



[Pranav Mistry]

The Great Challenge

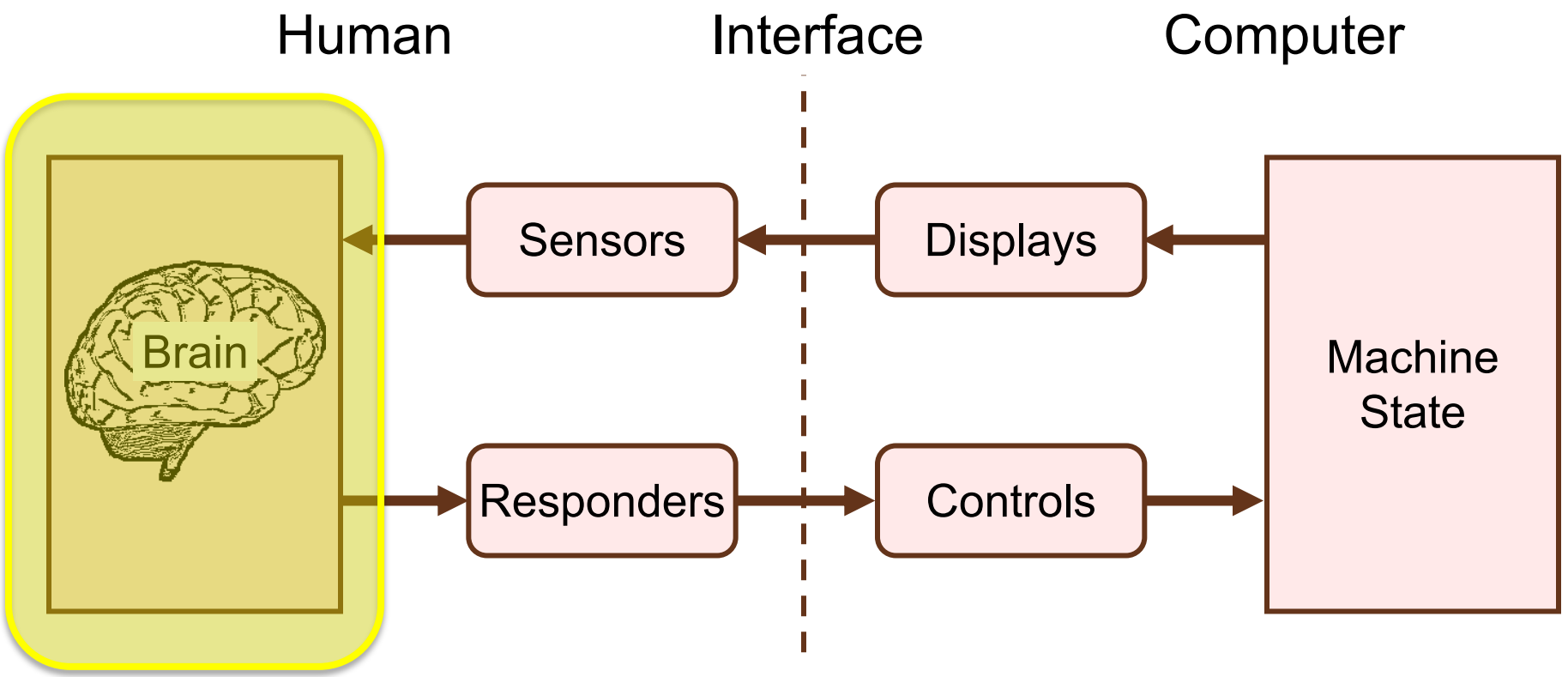
- Stephen Hawking - How was he talking?



[Speech Synthesizer] [Updated Software]

One more thing...

Human Factors Model



The Brain

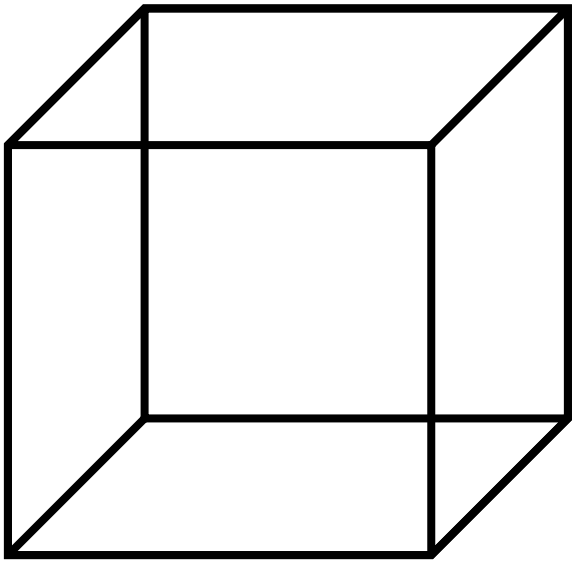
- Most complex biological structure known
- Billions of *neurons*
- Enables human capacity for…
  - Pondering, remembering, recalling, reasoning, deciding, communicating, etc.
- Sensors (human inputs) and responders (human outputs) are *connected* via the human brain
- Stages
  - **Perception, Cognition, Memory**

Perception [感知]

- First stage of processing for sensory input
- Associations with senses
  - Visual stimulus → familiar, strange
  - Auditory stimulus → harmonious, discordant
  - Smell stimulus → pleasurable, abhorrent
  - Taste stimulus → sweet, sour
  - Tactile stimulus → warm, hot

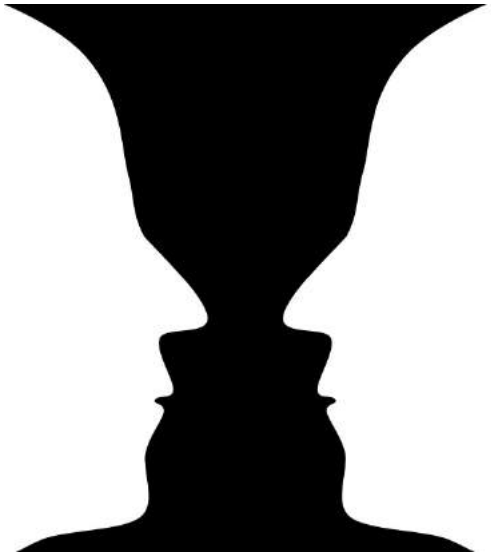
Ambiguity [歧义]

Necker cube [奈克方块]



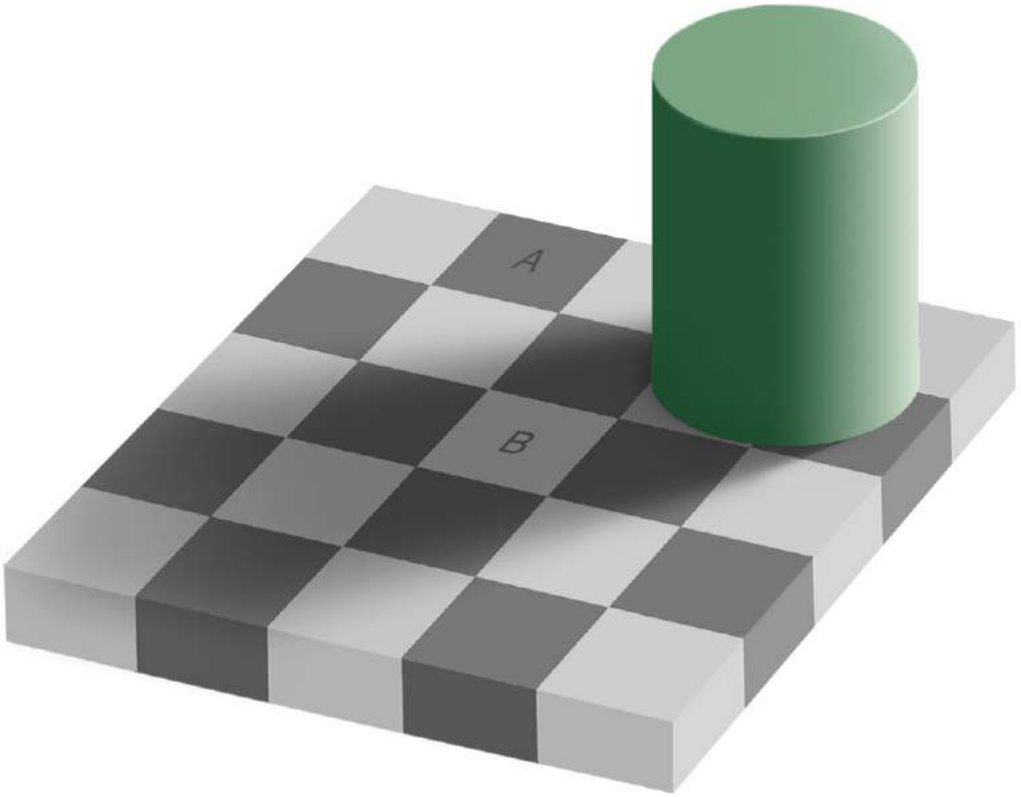
Which surface is at the front?

Rubin vase



Wine goblet or two faces?

Illusion [错觉]

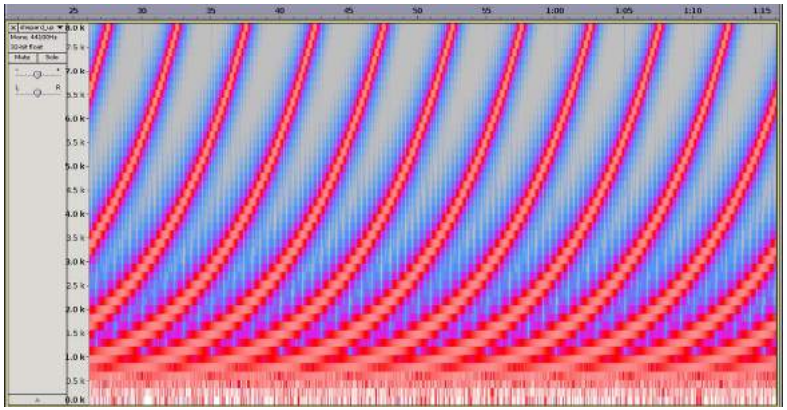


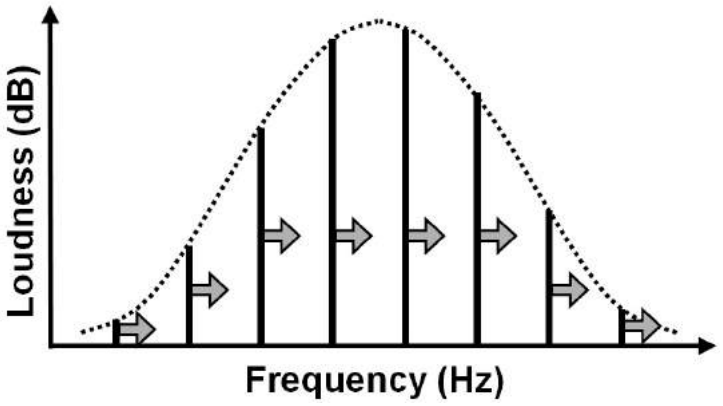
Are the colors of square A and B identical?

[Monument Valley]

Illusion – Other Senses

- If illusion is possible for the *visual* sense, the same should be true for the *other* senses
- Tactile/haptic illusion: [phantom limb] *[幻肢]*
- Auditory illusion: [Shepard-Risset glissando]

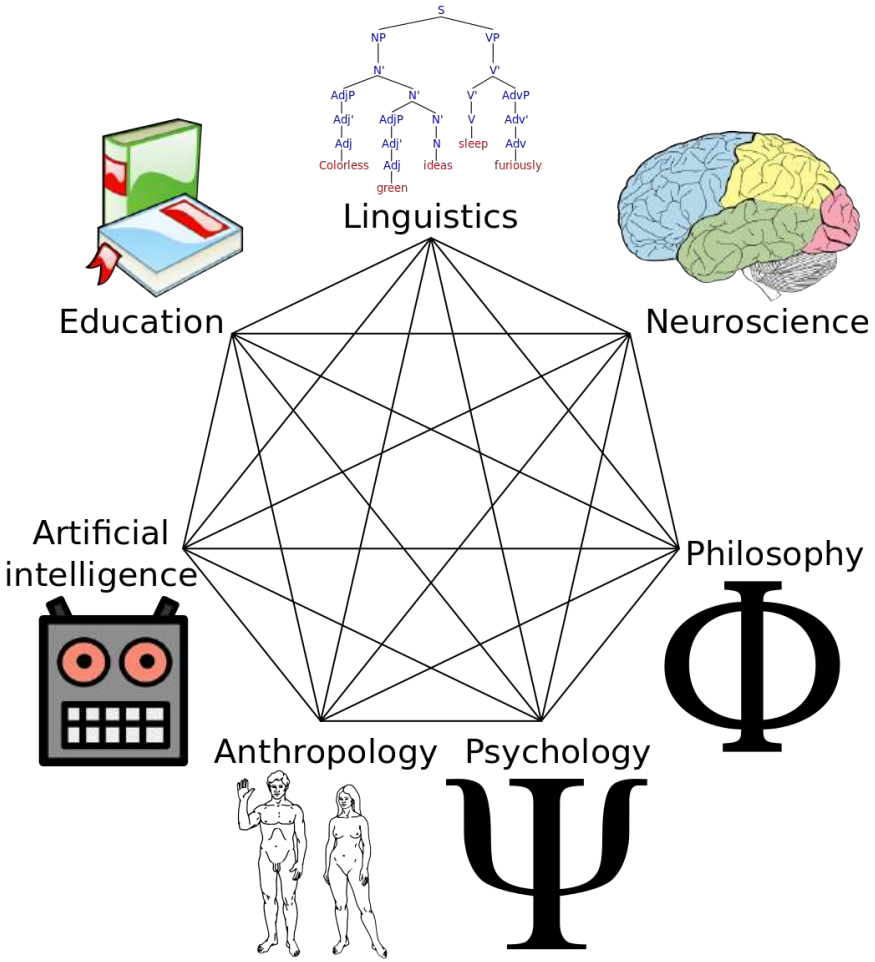




Cognition

- Cognition is the human process of *conscious intellectual* activity
  - Thinking, reasoning, deciding, etc.
- Spans many fields
  - Neurology, linguistics, anthropology, etc.
- Sensory phenomena → easy to study because they exist in the physical world
- Cognitive phenomena → hard to study because they exist *within* the human brain

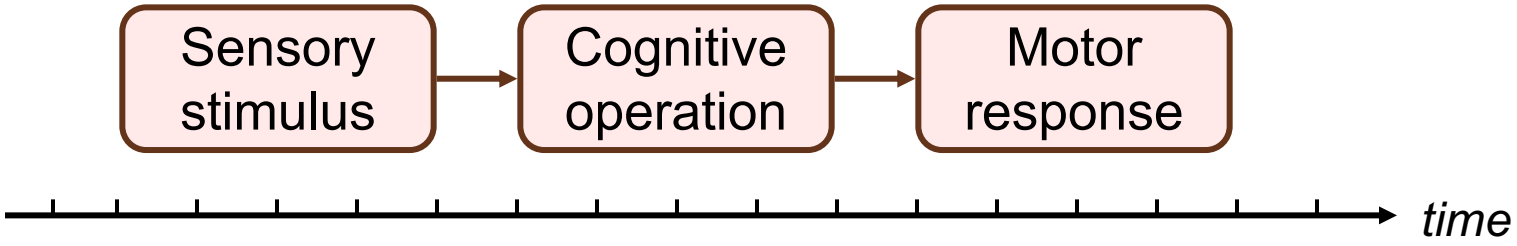
Cognition


S
NP
VP
N'
V'
AdjP
N'
V'
AdvP
Adj'
AdjP
N'
V
Adv'
Adj
Adj'
N
sleep
Adv
Colorless
Adj
ideas
furiously
green
Linguistics
Education
Neuroscience
Artificial intelligence
Philosophy
Anthropology
Psychology

“Making a Decision”

- Not possible to directly measure the time for a human to “make a decision”
  - *When* does the measurement begin and end?
  - *Where* is it measured?
  - On *what* input is the human deciding?
  - Through *what* output is the decision conveyed?
- There is a sensory stimulus and motor response that bracket the decision (next slide)

Making a Decision – in Parts



Operation	Typical time (ms)
Sensory reception	1 – 38
Neural transmission to brain	2 – 100
Cognitive processing	70 – 300
Neural transmission to muscle	10 – 20
Muscle latency and activation	30 –70
Total:	113 - 528

Large variation!

Examples of Simple Decisions

- Driving a car → decision to depress the brake pedal in response to a changing signal light
- Using a mobile phone → decision to press REJECT-CALL in response to an incoming call
- Reading news online → decision to click the CLOSE button on a popup ad
- These are *reaction time* tasks (discussed shortly)

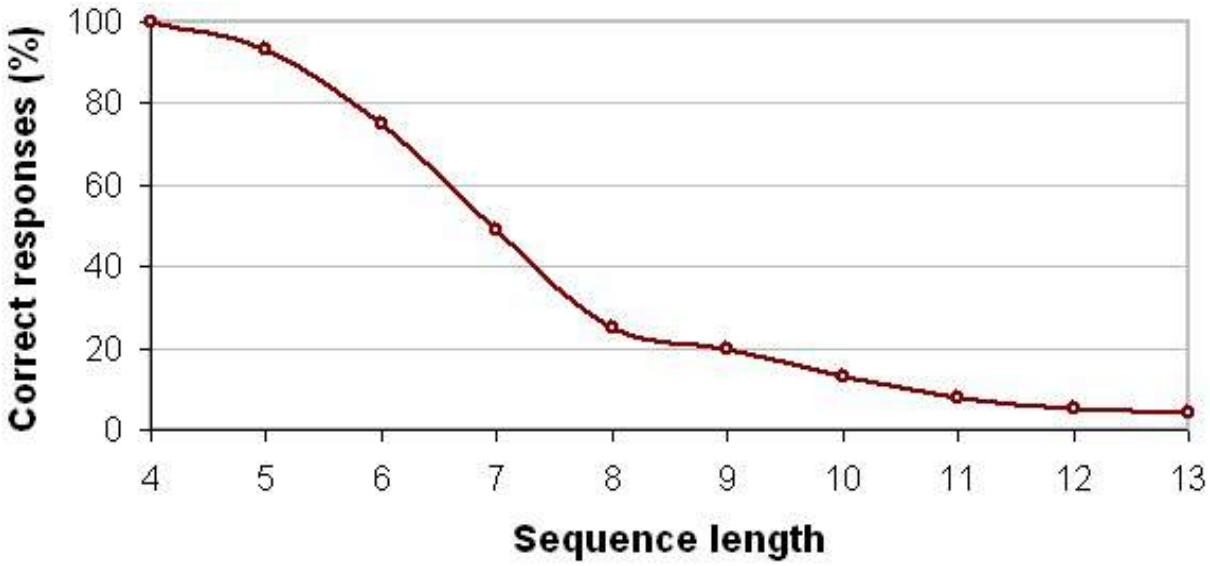
Memory

- Vast repository
- Long-term memory
  - Declarative/explicit area → information about events in time and objects in the external world
  - Implicit/procedural area → information about how to use objects and how to do things
- Short-term memory
  - Aka *working memory*
  - Information is active and readily available for access
  - Amount of working memory is small, about 7 (±2) units or chunks[1]

[1] Miller, G. A. (1956). The magical number seven plus or minus two: Some limits on our capacity for processing information. *Psychological Review, 63*, 81-97.

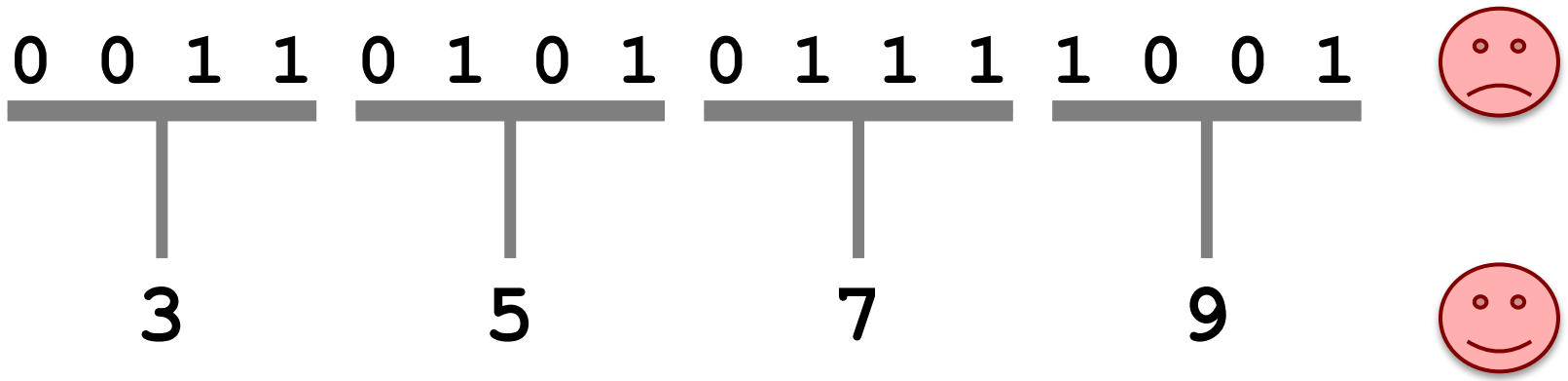
Short Term Memory Experiment

- Random sequences of digits recited to subjects
- Sequences vary from 4 to 13 digits
- After recitation, subjects copy sequence from memory to a sheet of paper
- Transcriptions on sheets scored (correct/incorrect)
- Results ($n \approx 60$):

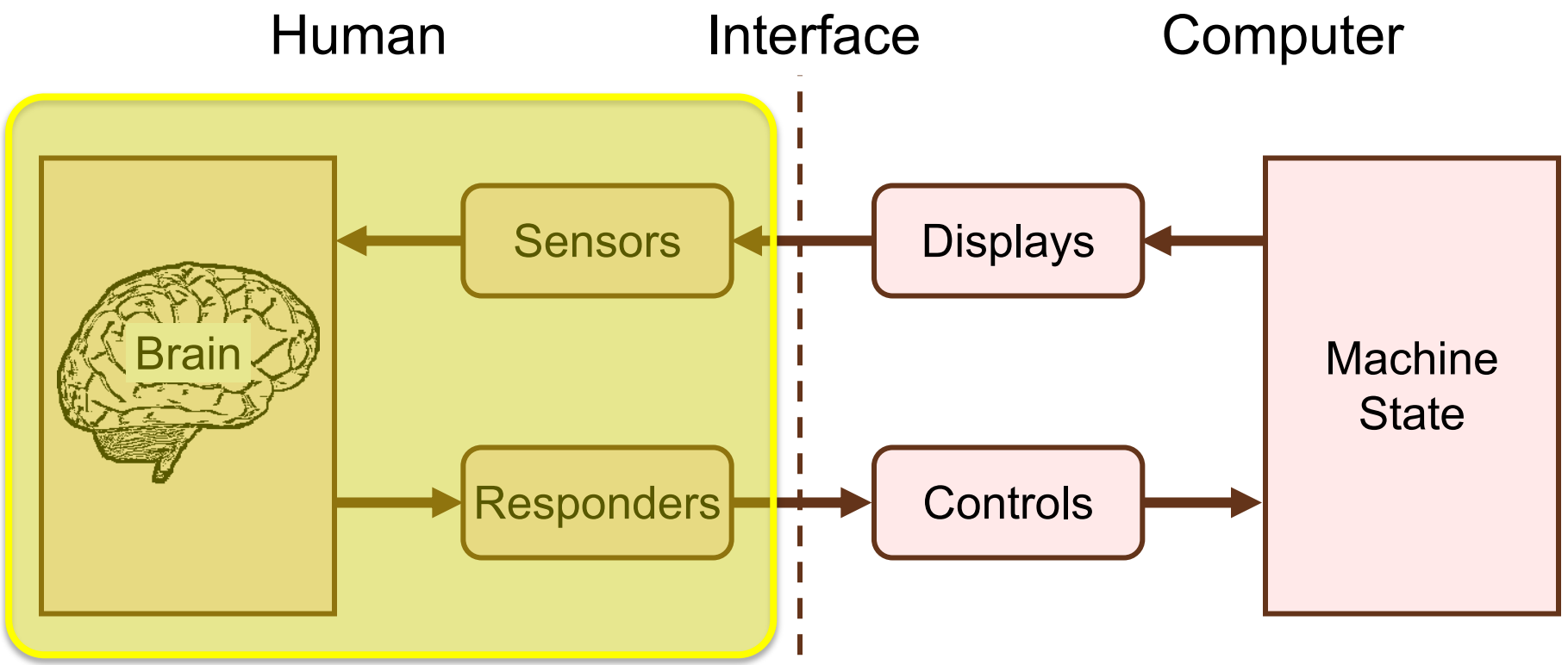


Chunking

- Units in short term memory may be recoded as a chunk
- Expands capacity of short term memory
- E.g., Commit to memory and recall…



Human Performance



Skilled Behaviour
Attention
Human Error

Human Performance

- Humans uses their sensors, brain, and responders to do things
- When the three work together to achieve a goal, human performance arises
- Examples:
  - Tying shoelaces
  - Folding clothes
  - Searching the web
  - Entering a text message on a mobile phone

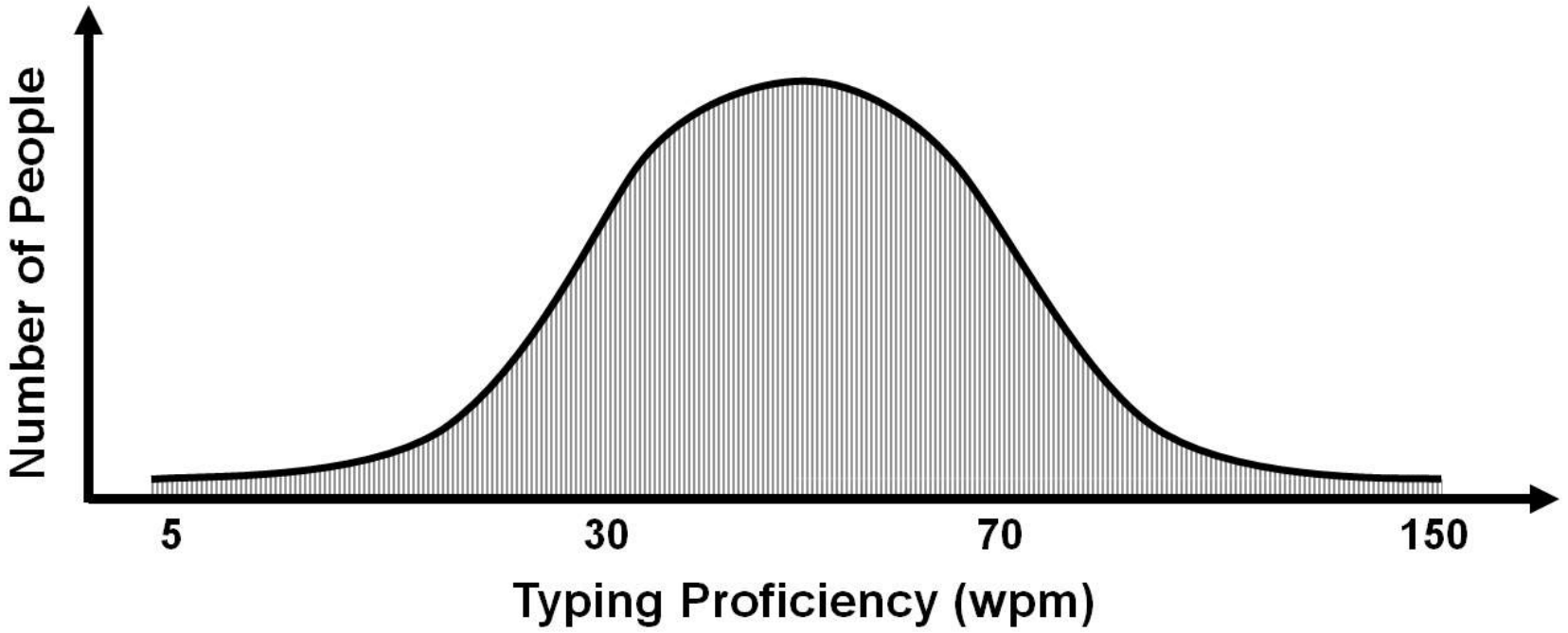
Speed-Accuracy Trade-off

- Fundamental property of human performance
- Go faster and errors increase
- Slow down and accuracy improves
- HCI research on a new interface or interaction technique must consider both the **speed** in doing tasks and the accompanying **accuracy**

Human Diversity

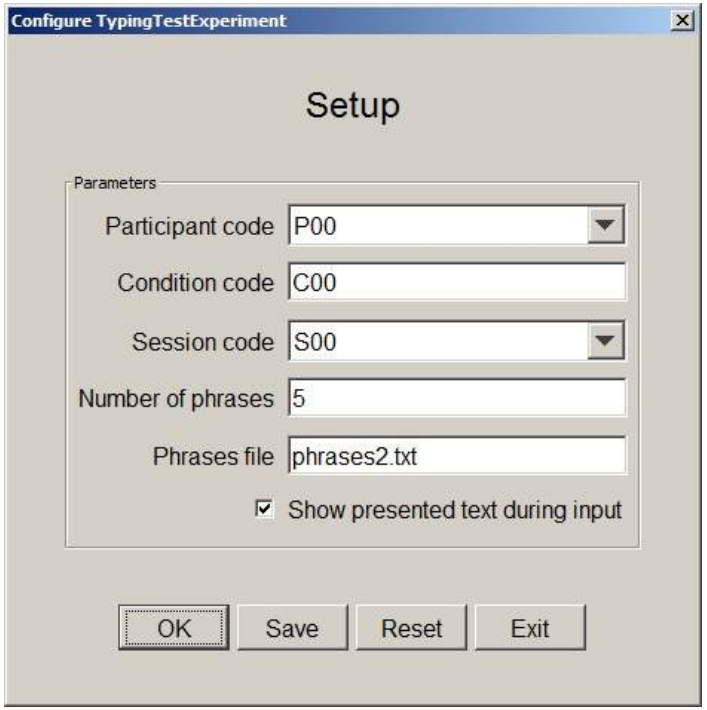
- Human performance is highly complex:
  - Humans differ (age, gender, skill, motivation, etc.)
  - Environmental conditions affect performance
  - Secondary tasks often present
- Human diversity and human performance often shown in a distribution (next slide)

Human Diversity and Performance

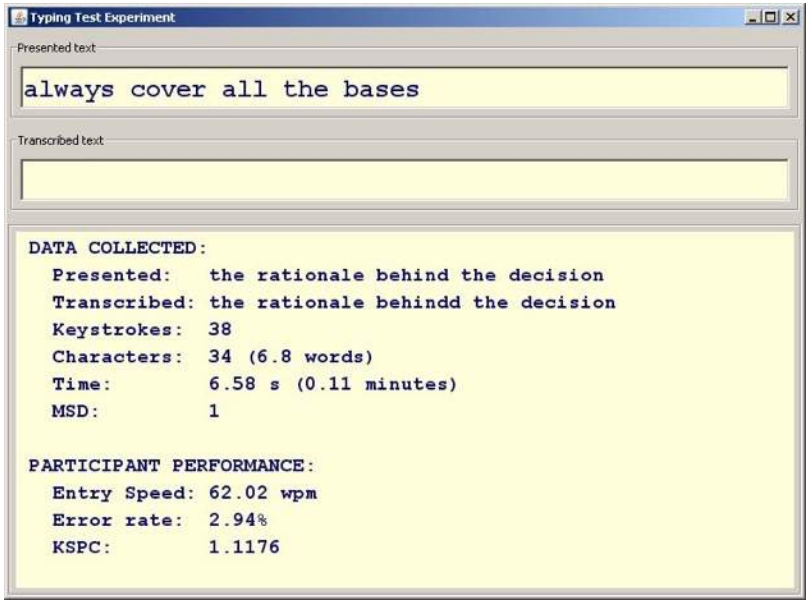


Where are you on this chart?
Where is your mother or grandmother?
Where is an 8-year old, just learning to use a computer?
Where is someone with a physical disability?
Where are you while using your mobile phone on a crowded bus (standing!)?

Typing Test Experiment







Use software from book's web site:

http://www.yorku.ca/mack/HCIbook/

Reaction Time

- One of the most primitive manifestations of human performance is *simple reaction time*
- Definition: The delay between the occurrence of a single fixed stimulus and the initiation of a response assigned to it[1]
- Example: pressing a button in response to the onset of a stimulus light

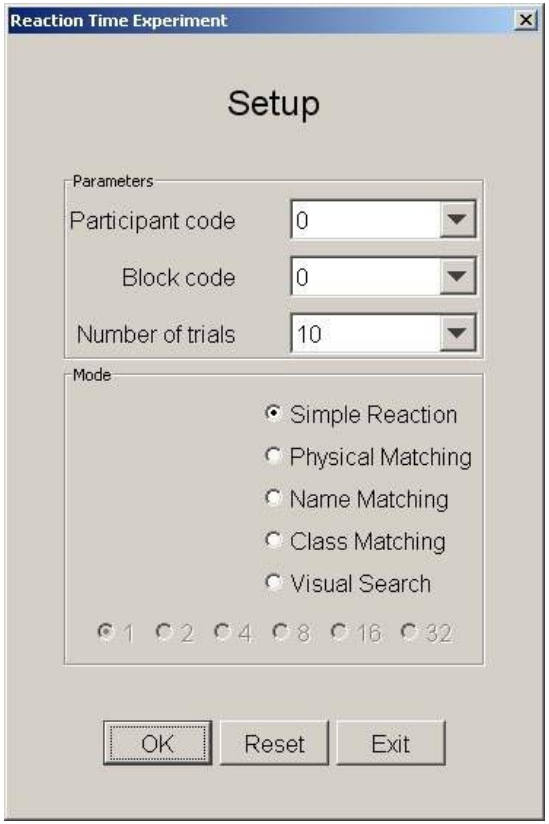
[1] Fitts, P. M., & Posner, M. I. (1968). *Human performance*. Belmont, CA. Brooks/Cole Publishing Company.

Sensory Stimuli and Reaction Time

- Delay time varies by type of sensory stimuli
- Approximate values[1]
  - Auditory → 150 ms
  - Visual → 200 ms
  - Smell → 300 ms
  - Pain → 700 ms

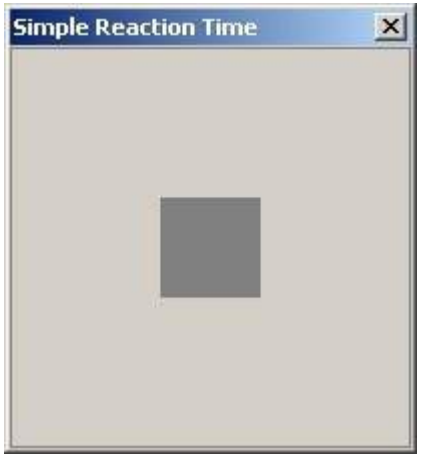
[1] Bailey, R. W. (1996). *Human performance engineering: Designing high quality, professional user interfaces for computer products, applications, and systems* (3rd ed.). Upper Saddle River, NJ: Prentice Hall.

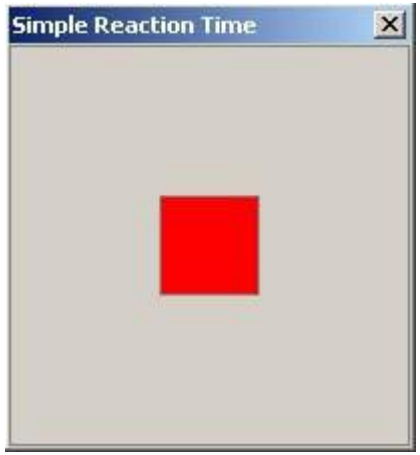
Reaction Time Experiment





Simple Reaction Time



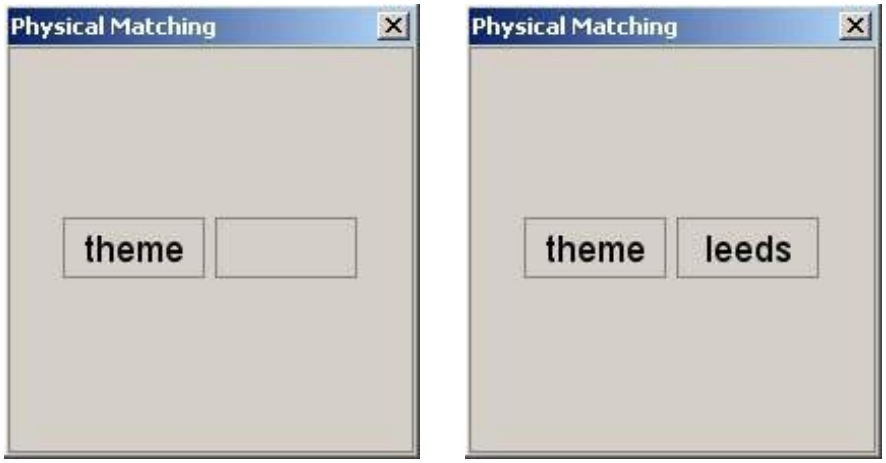


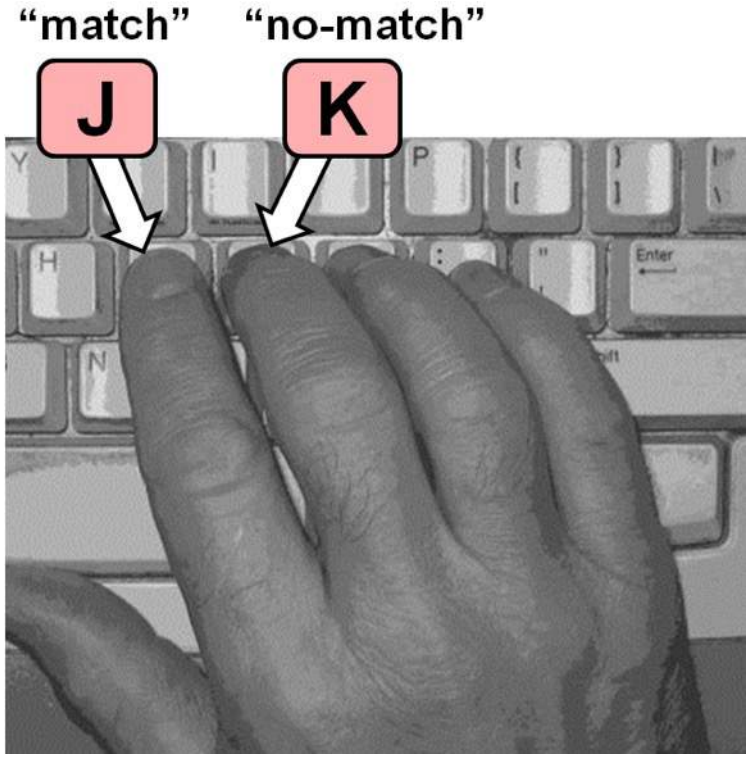
Use software from book's web site:

http://www.yorku.ca/mack/HCIbook/

Reaction Time Experiment (2)

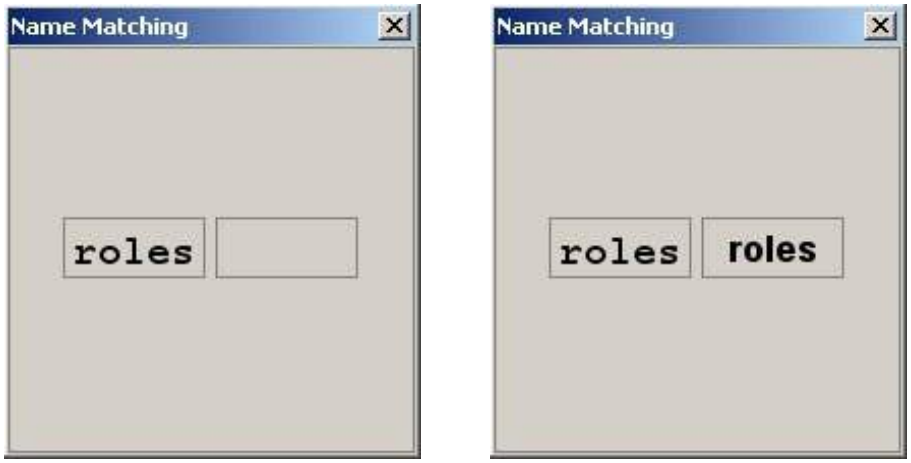
Physical Matching



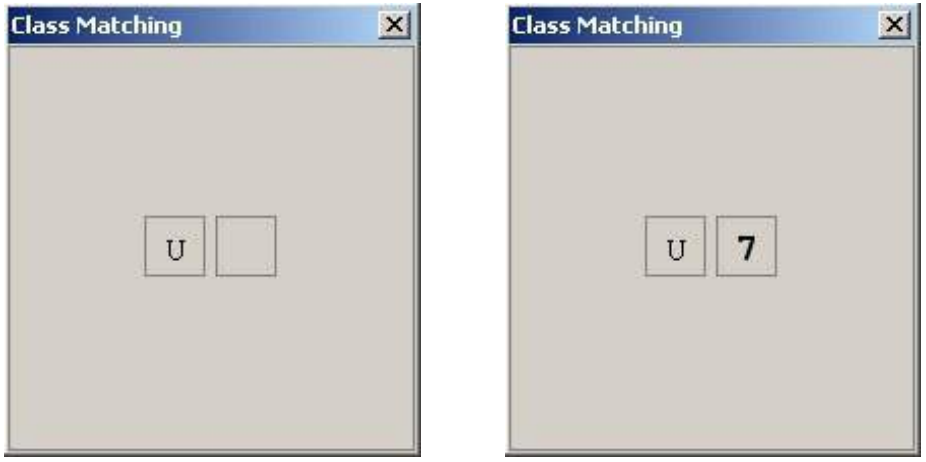


Reaction Time Experiment (3)

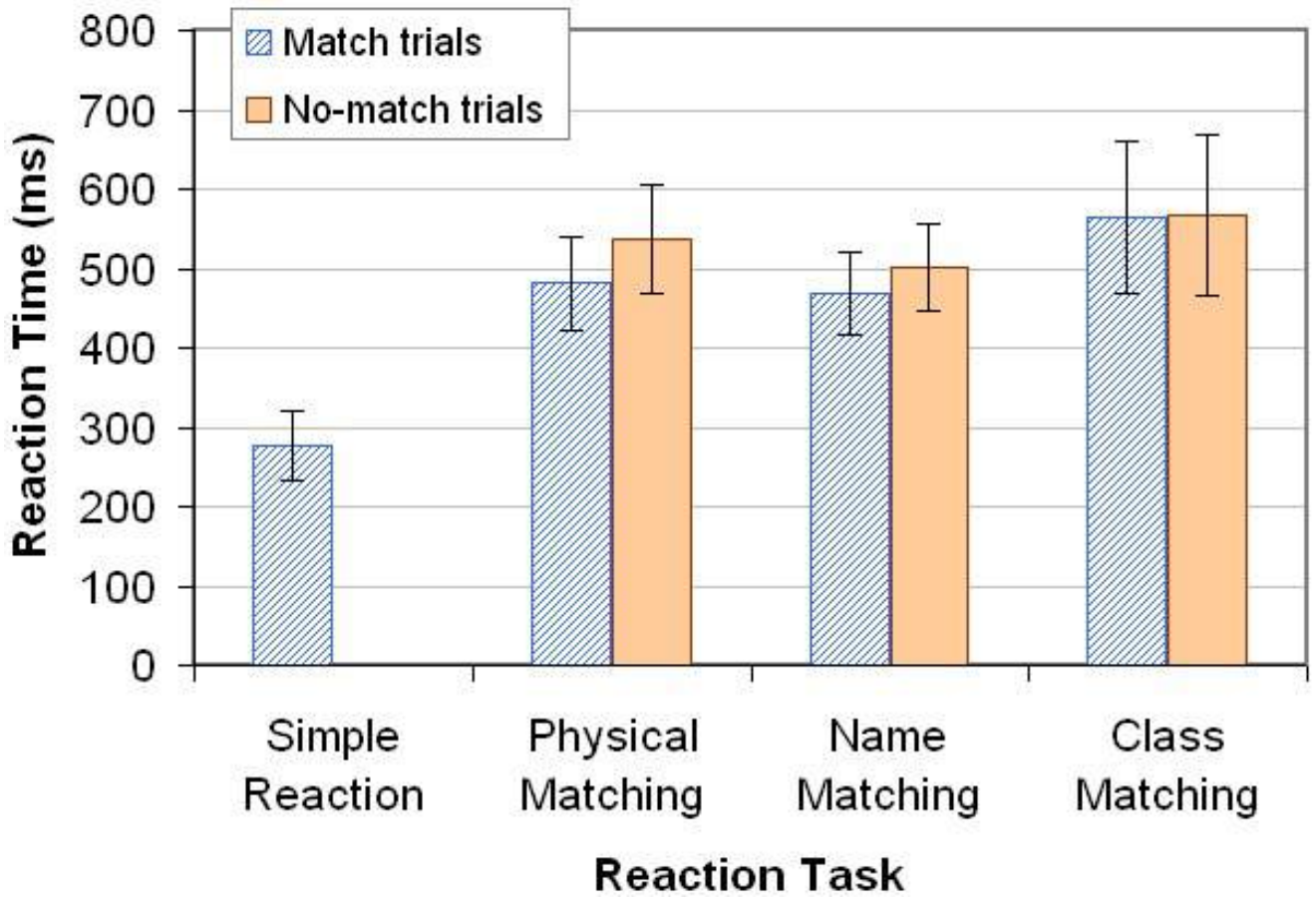
Name Matching



Class Matching

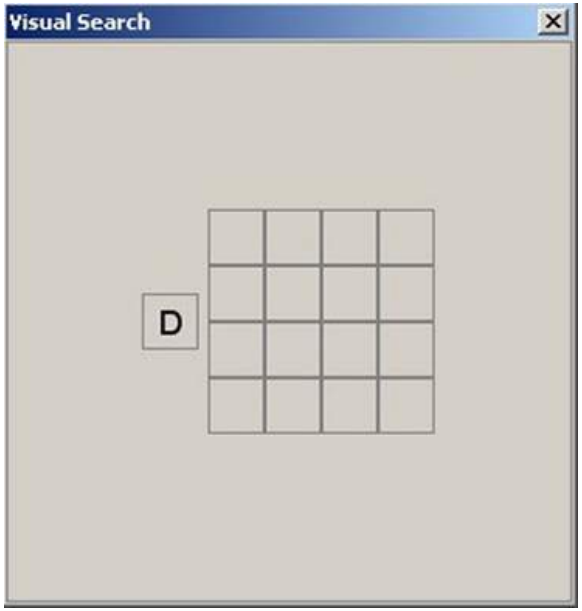


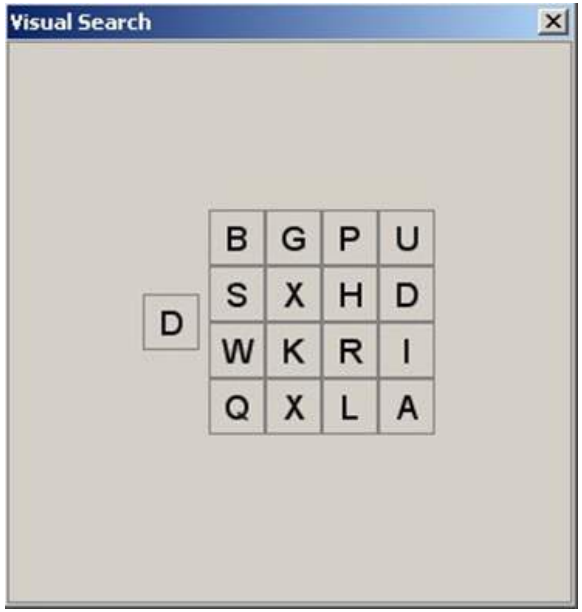
Experiment Results



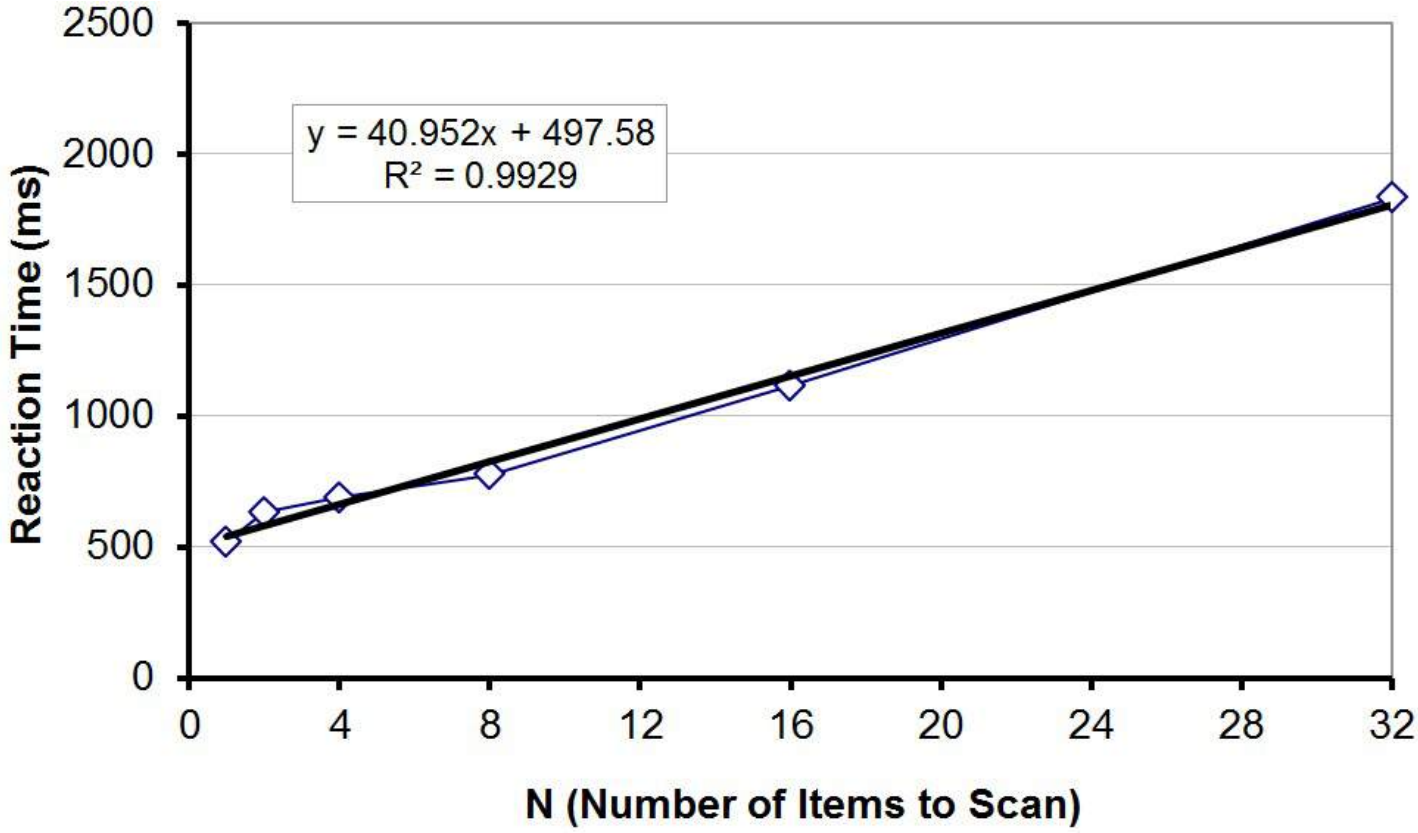
Visual Search

- A variation on simple reaction time
- User scans a collection of items looking for desired item
- Time increases with the number of items to scan
- Included in the demo software with $N = 1, 2, 4, 8, 16$, or 32 items

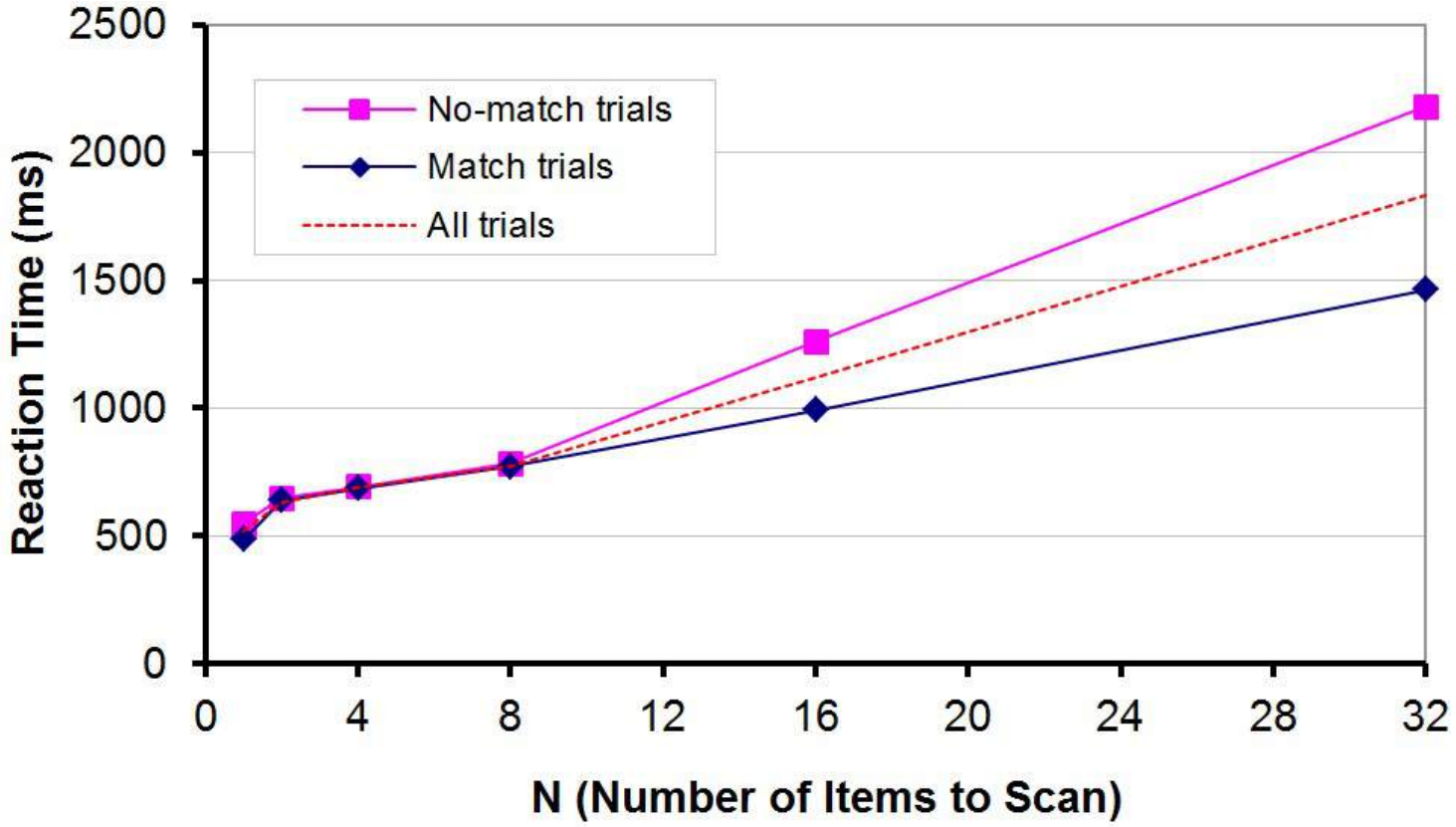




Experiment Results (1)



Experiment Results (2)



Skilled Behaviour

- For many tasks, human performance improves considerably and continuously with practice
- (Note: Very little improvement with practice in simple reaction time tasks)
- In these tasks, there is interest in studying the progression of learning and the performance achieved according to the amount of practice
- Categories of skilled behavior:
  1. Sensory-motor skill (e.g., darts, gaming)
  2. Mental skill (e.g., chess, programming)
  - Some tasks required a lot of both (next slide)

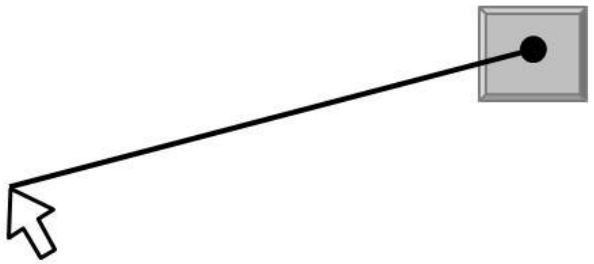
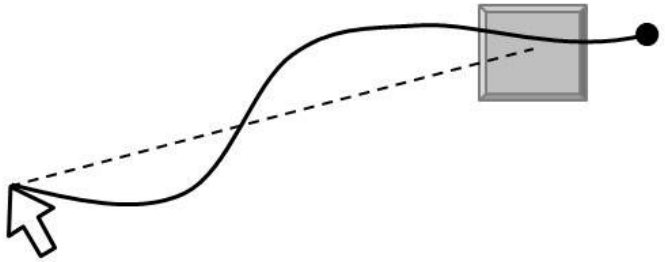
Attention

- *Texting while driving!*
- Attention is complex:
  - Divided attention, secondary tasks
  - Which tasks require attention?
    - Can't talk and type
  - Which tasks do not require attention?
    - Can talk and walk
  - What are the human limits? *What is attention?*
- Two categories of attention
  - Divided attention (attending to more than one task)
  - Selected attention (attending to one task to the exclusion of others)

Human Error

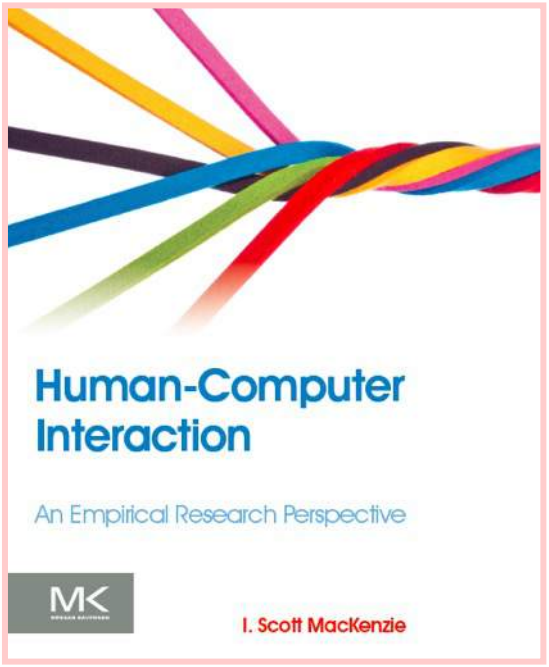
- Human error can be studied from many levels
- Simple view: An error is a discrete event where the outcome deviated from the desired outcome
- But, tasks in error are often at least partly correct (next slide)

Variability and Error

	Target Selection	Text Entry
Correct		quickly
Incorrect		qucehkly

What went wrong and why?

Thank You


Human-Computer Interaction
An Empirical Research Perspective
MK
I. Scott MacKenzie